

# The modern MLOps framework

MLOPS DEPLOYMENT AND LIFE CYCLING



**Nemanja Radojkovic**  
Senior Machine Learning Engineer

# About me

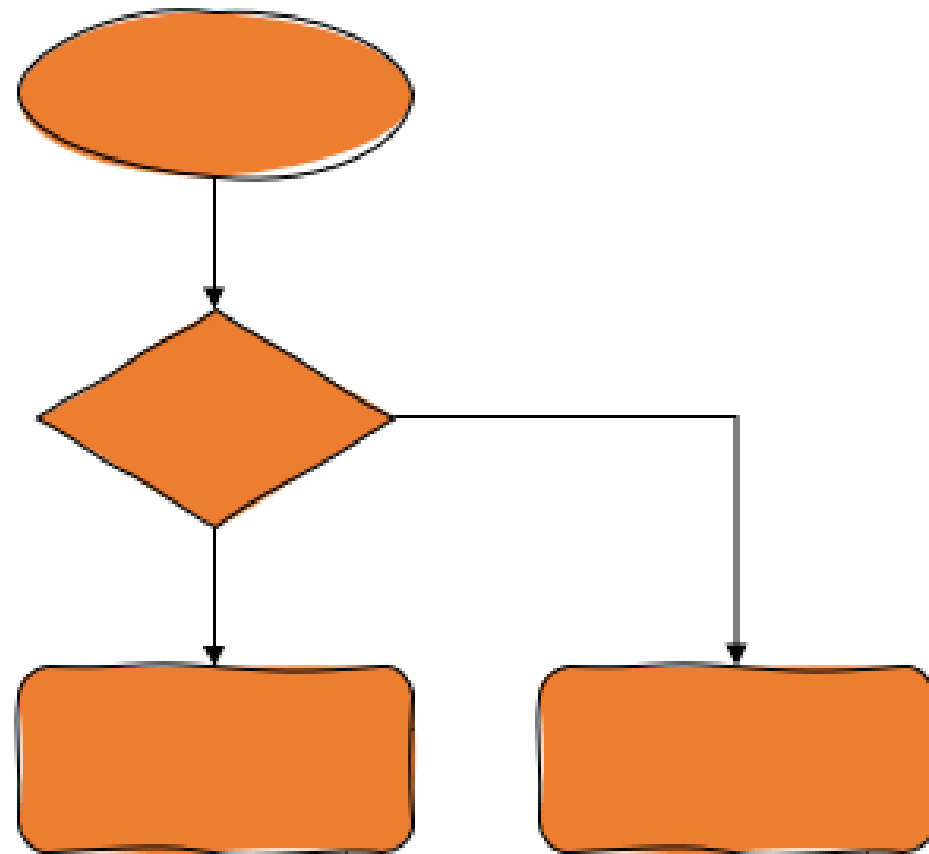


- Senior Machine Learning Engineer
- Experience developing, deploying, and maintaining Machine Learning models in production
- Current focus: ML deployment and life-cycling

<sup>1</sup> [www.linkedin.com/in/radojkovic](https://www.linkedin.com/in/radojkovic)

# Conceptual course

YES: High-level understanding



NO: Hands-on skills



# MLOps

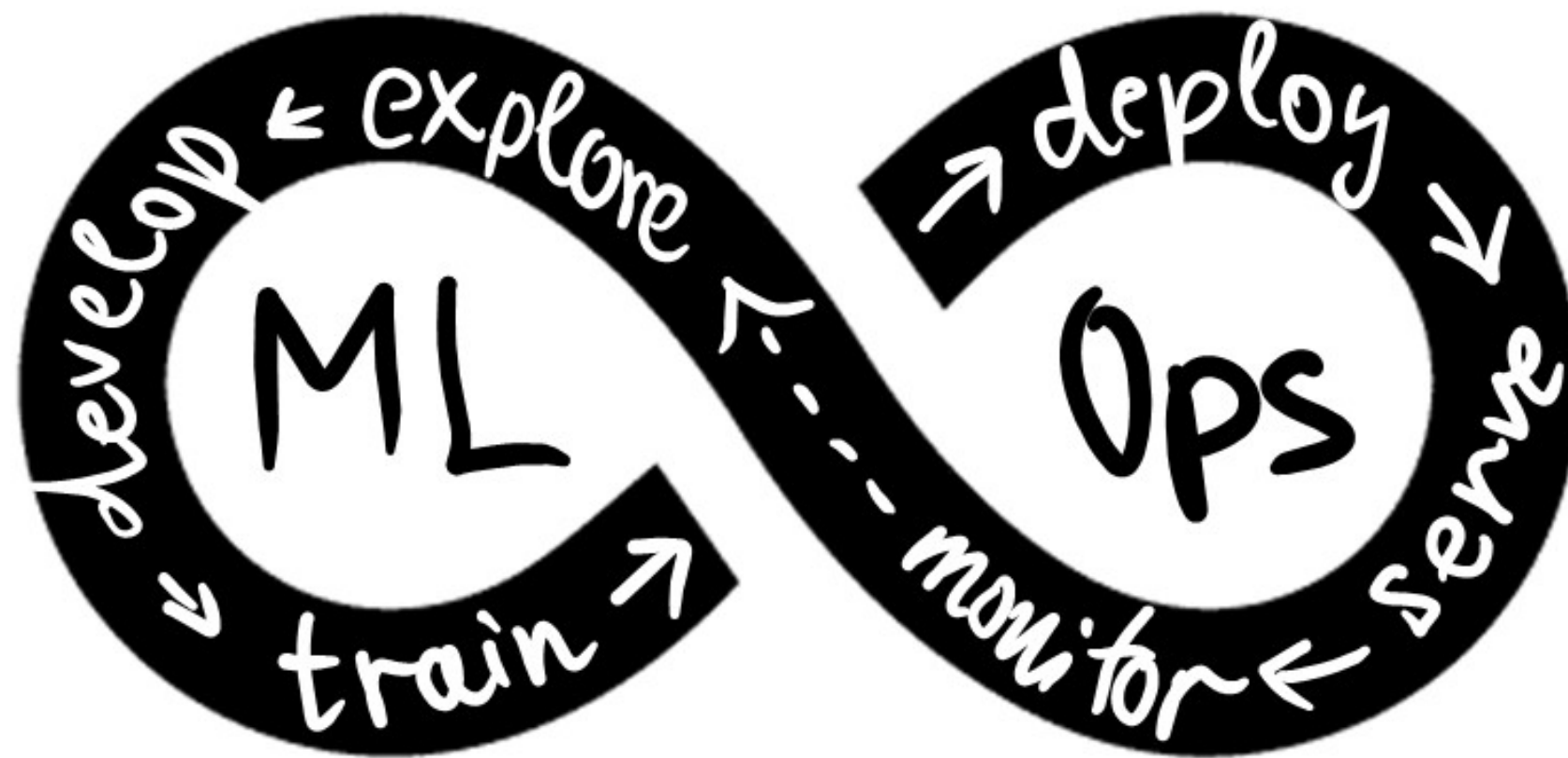
## Machine Learning Operations

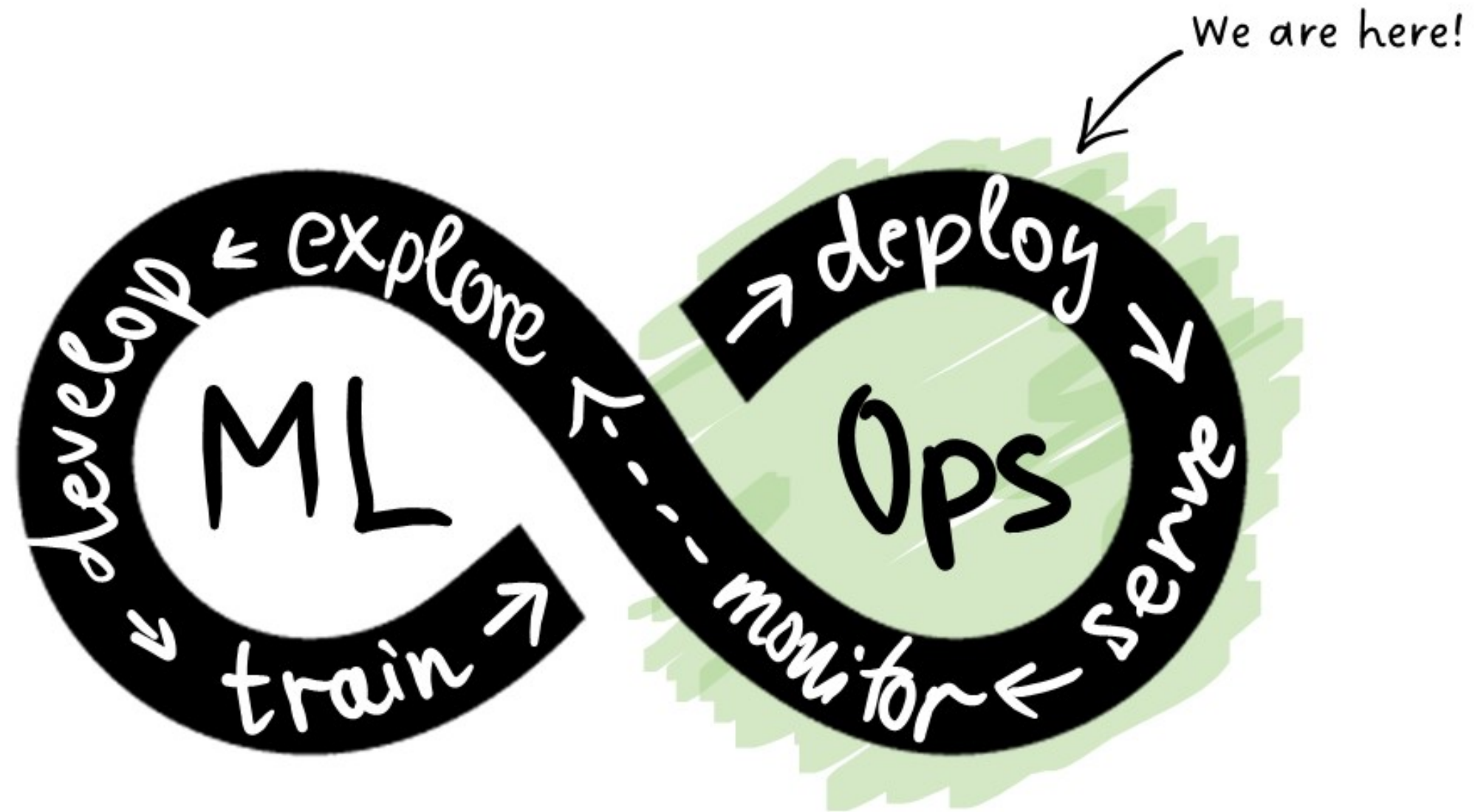
- Principles
- Practices
- Tools

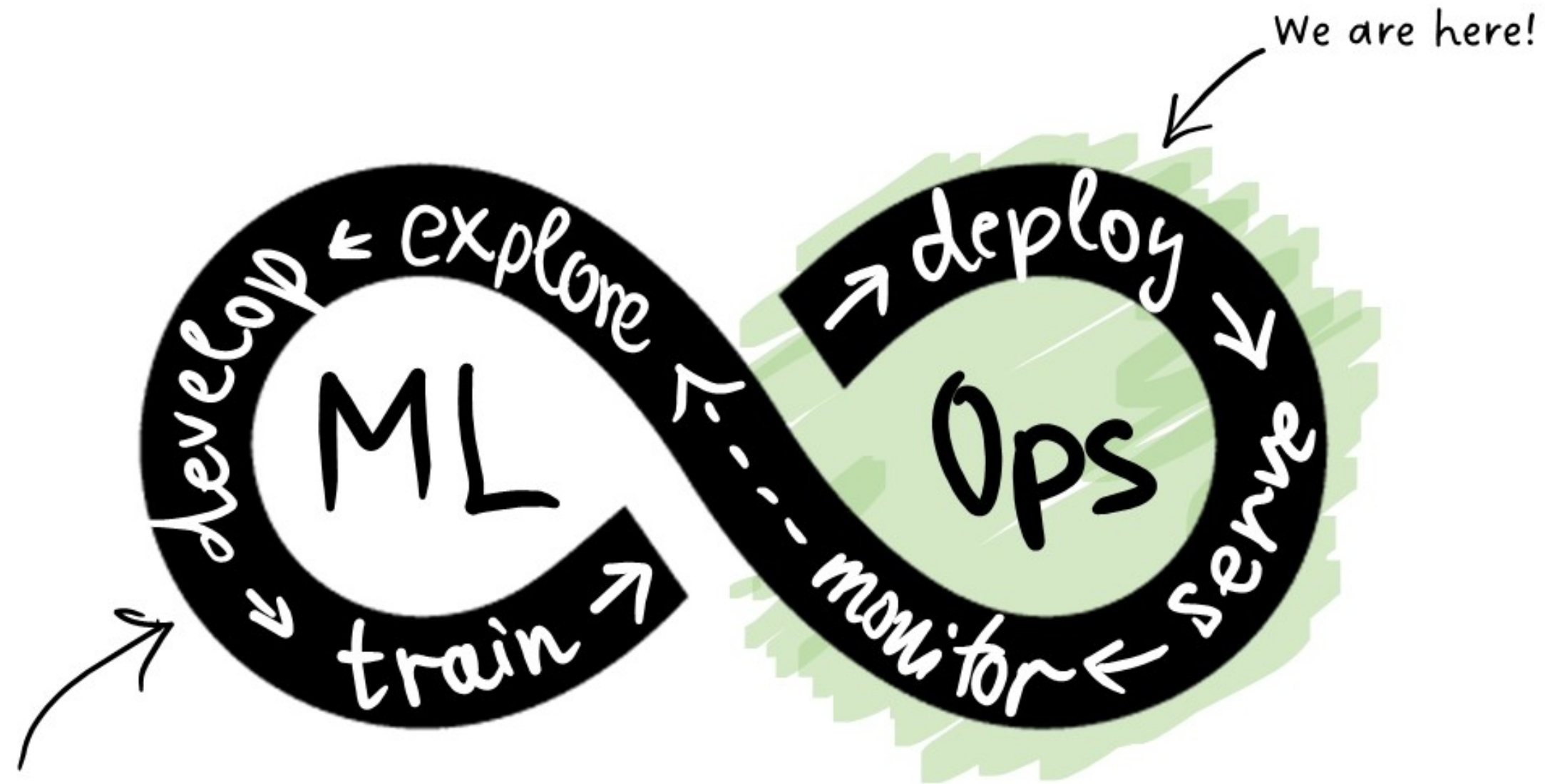
## Goal: ML workflows and services

- Automated
- Reproducible
- Integrated









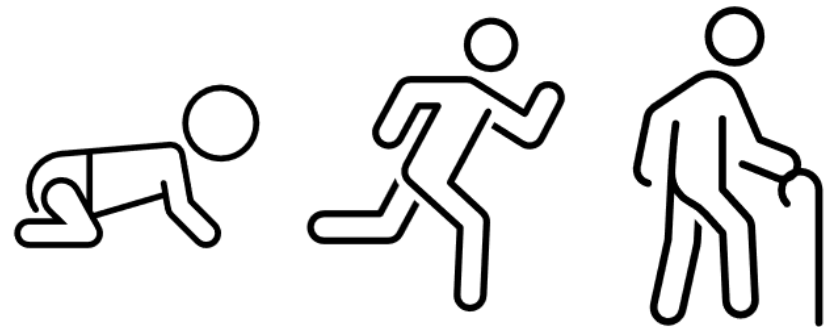
But! Never forget that  
MLOps starts here!

# Chapter I

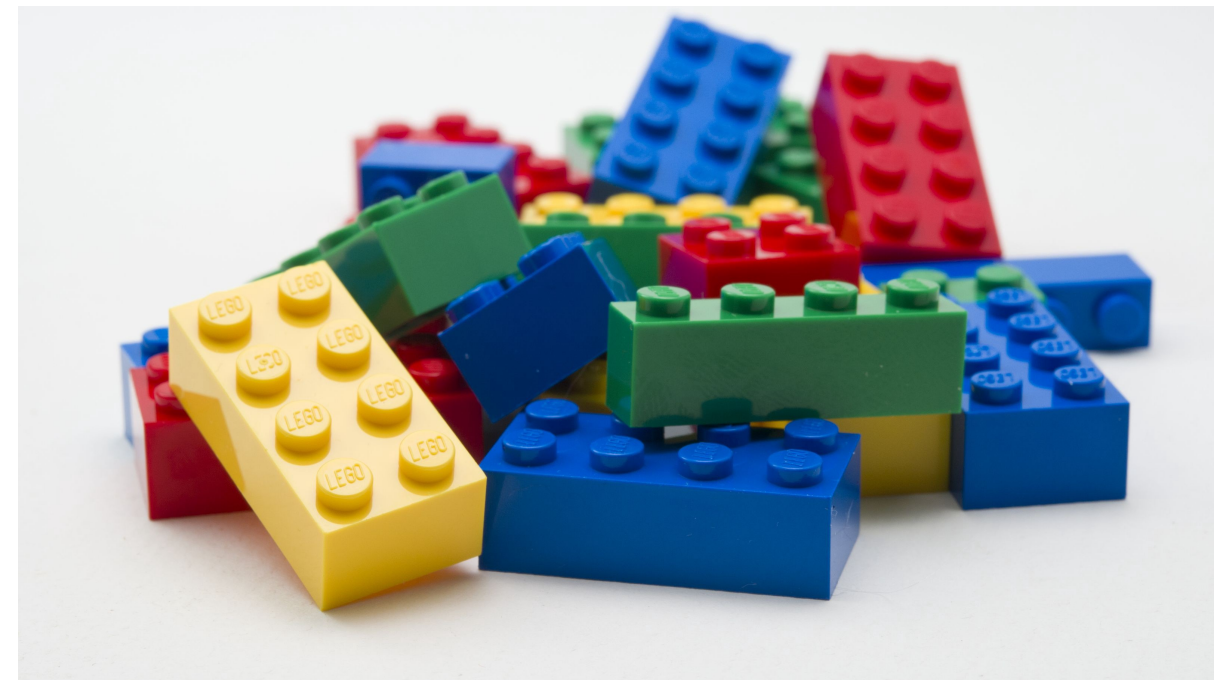
## Value and necessity of MLOps

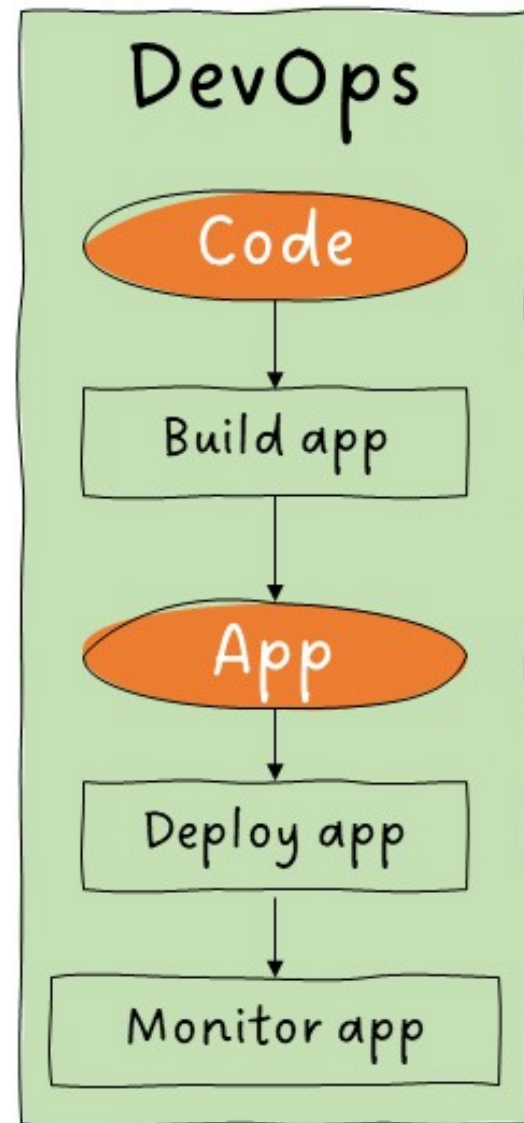


## Model life cycle stages

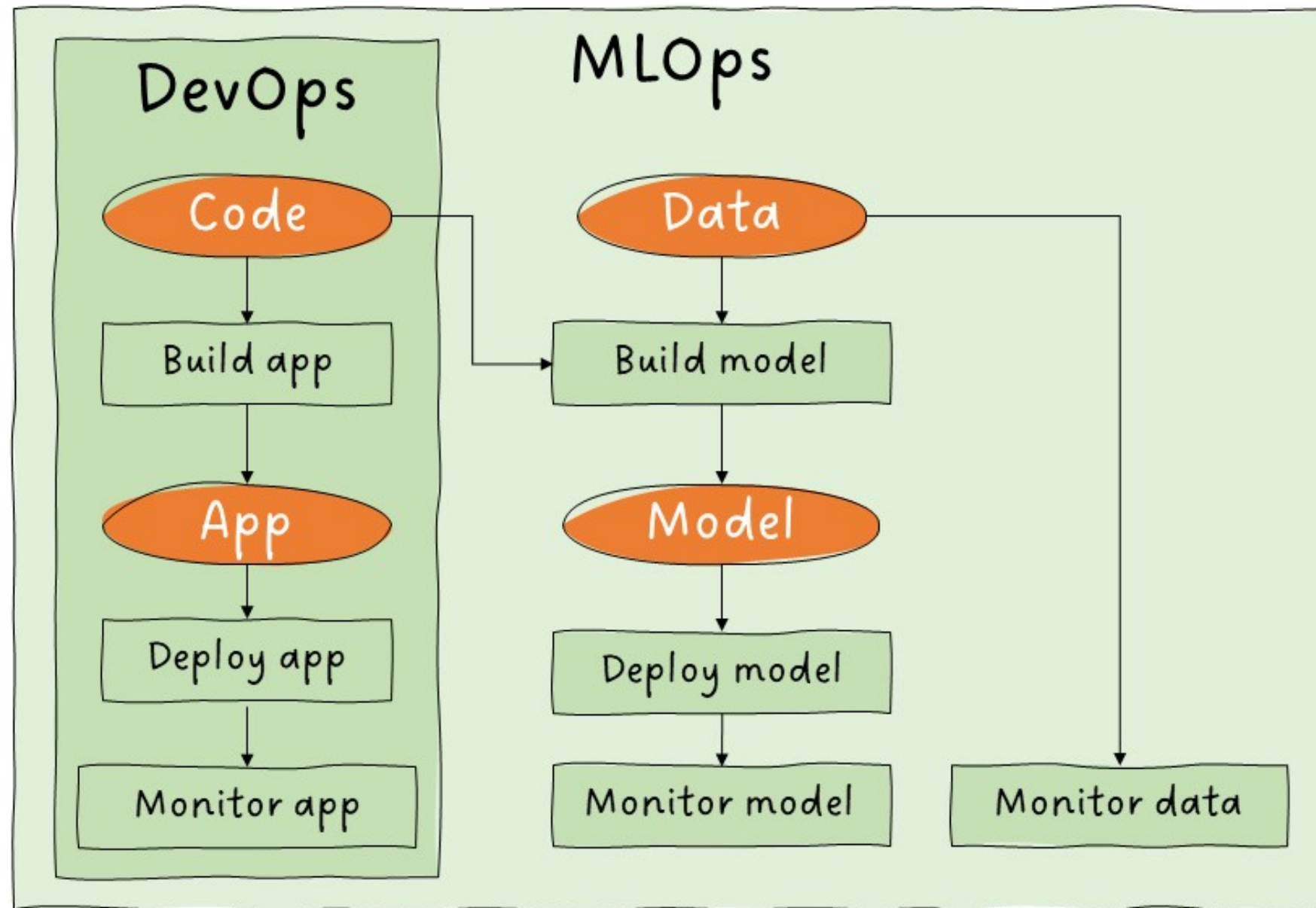


## Components of the MLOps framework











# High cost of ML without the Ops

## USUAL STARTING POINT

- Manual ML workflows
- Manual deployment
- Ad hoc monitoring

=> Accumulation of technical debt

# Technical Debt

the implied cost of additional rework caused by choosing an easy (limited) solution now instead of using a better approach that would take longer

~ *Wikipedia*[1]

**Famous Google paper on the topic:**

*"Machine Learning: The high-interest credit card of technical debt"*[2]

## MORE TIME AND MODELS DEPLOYED

- More technical debt and model risk
- Process slower, more frustrating and error prone
- Ability to deliver value impaired

<sup>1</sup> [https://en.wikipedia.org/wiki/Technical\\_debt](https://en.wikipedia.org/wiki/Technical_debt) <sup>2</sup> <https://research.google/pubs/pub43146/>

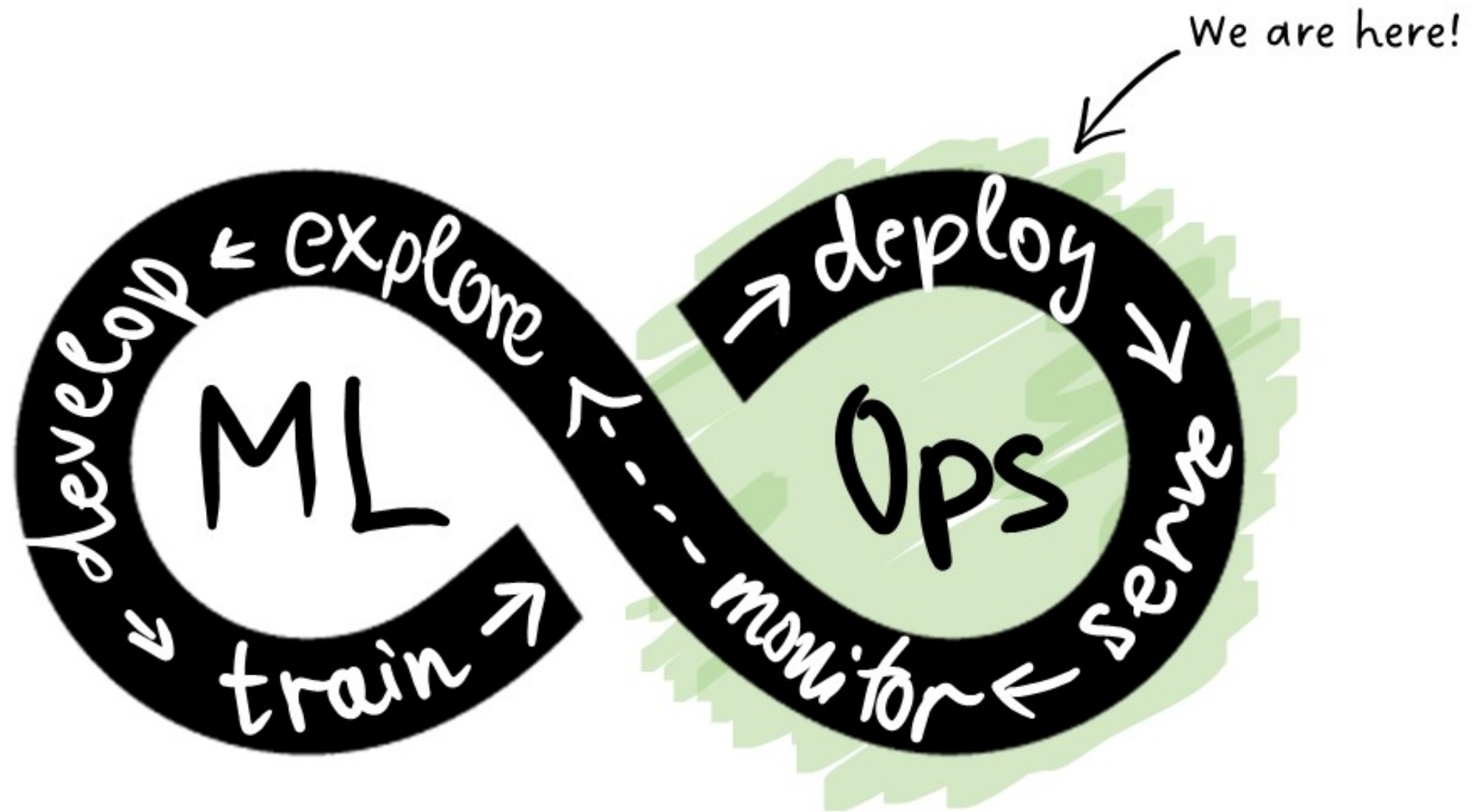
# ML workflows

- Data collection and preparation
- Data-labeling
- Model selection
- Model training
- Model packaging
- Model deployment
- Model monitoring and maintenance

ML workflow automation == MLOps maturity

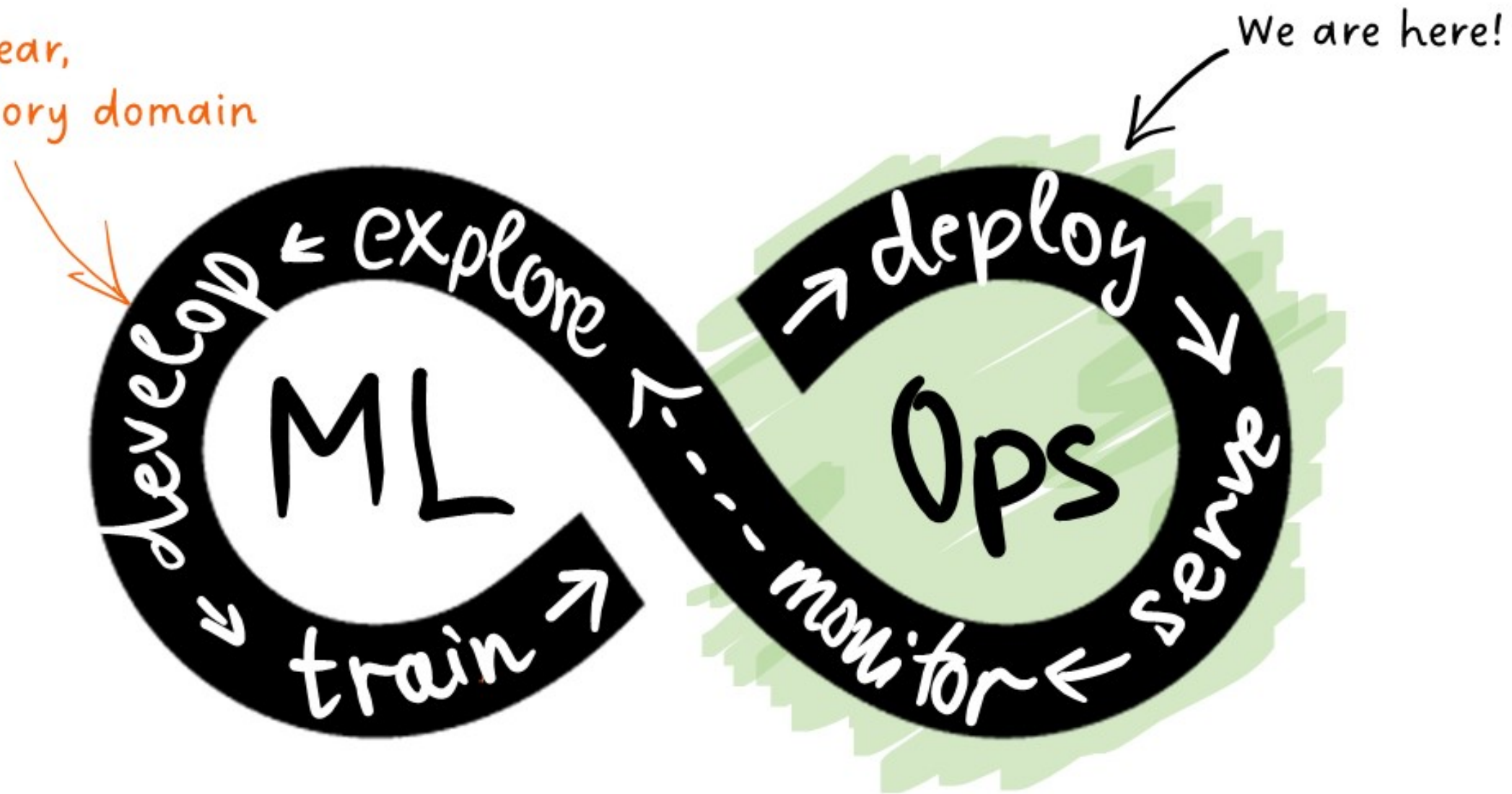
# MLOps implemented

- Automated
- Fast
- Reproducible
- Explainable

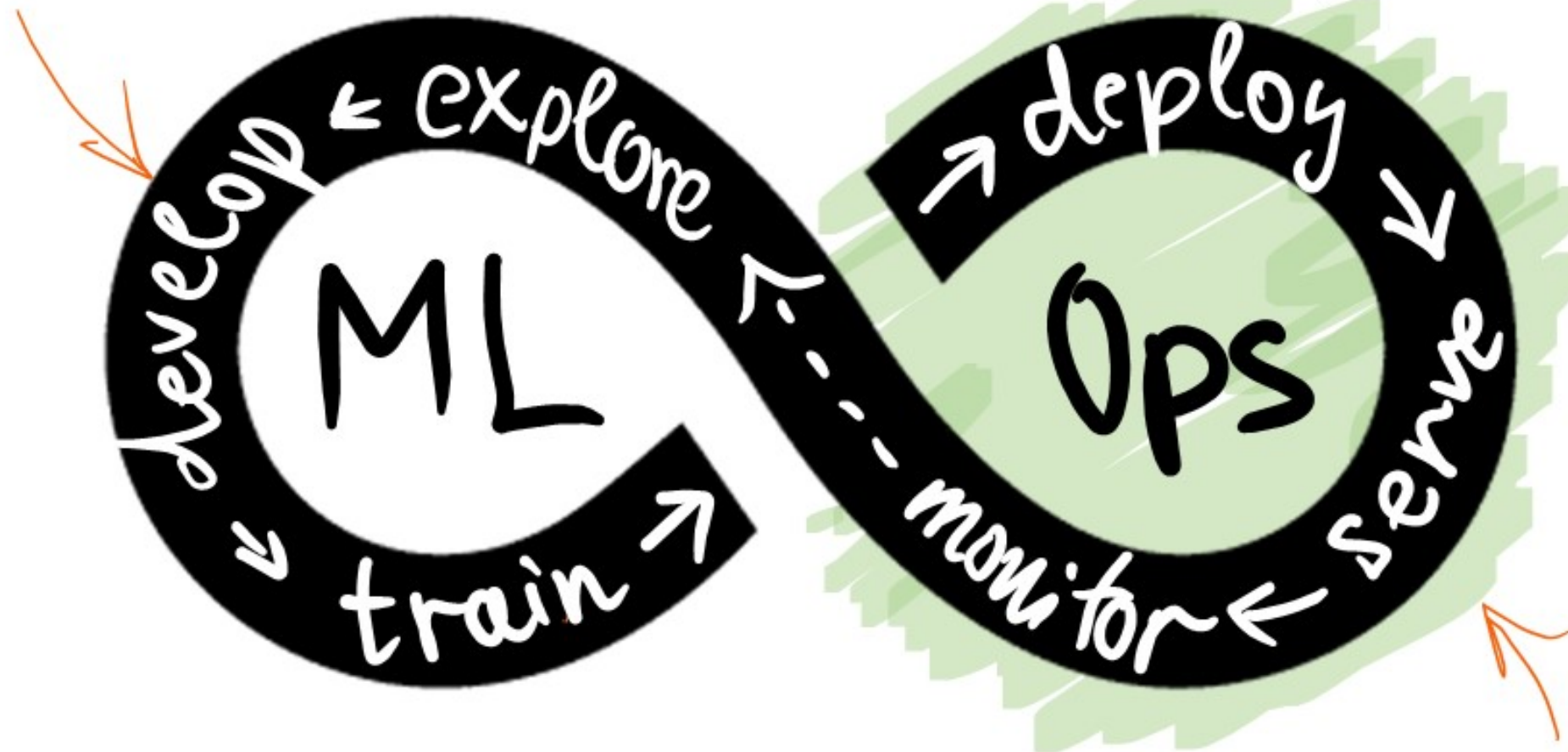




non-linear,  
exploratory domain

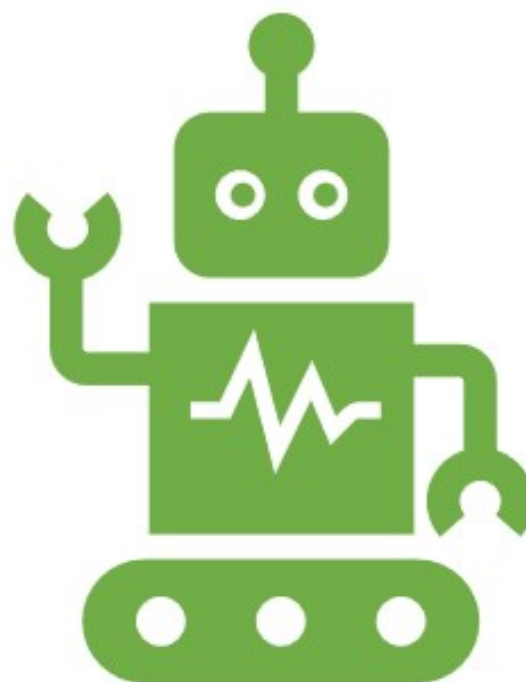


non-linear,  
exploratory domain



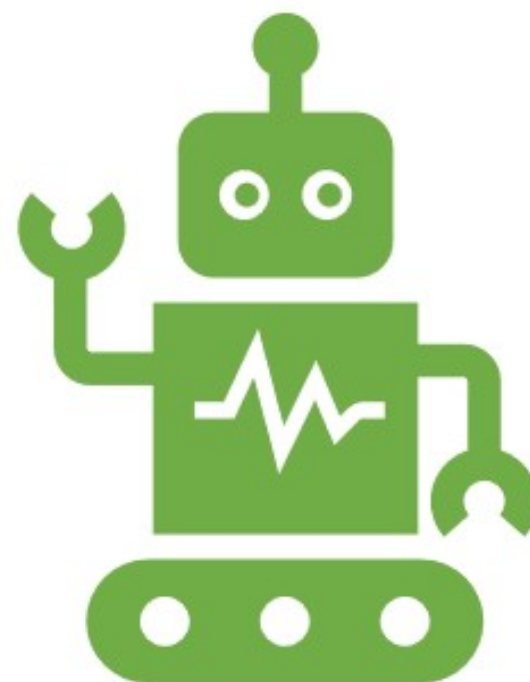
We are here!

streamlined,  
structured domain





high visibility



short  
response  
time

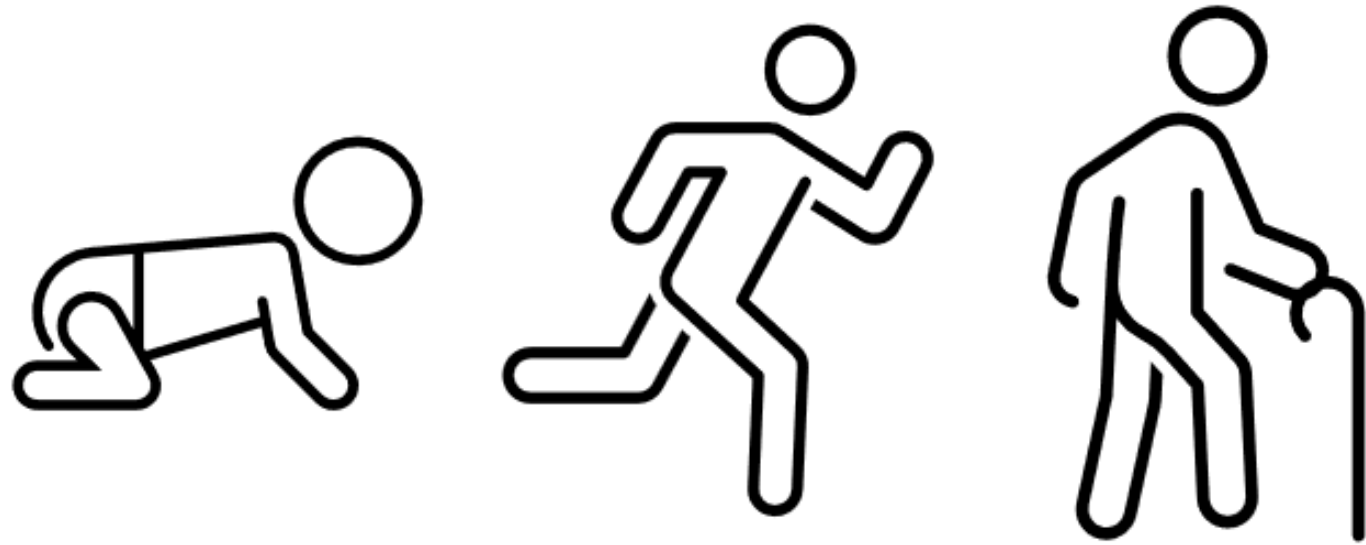
narrow error  
margins



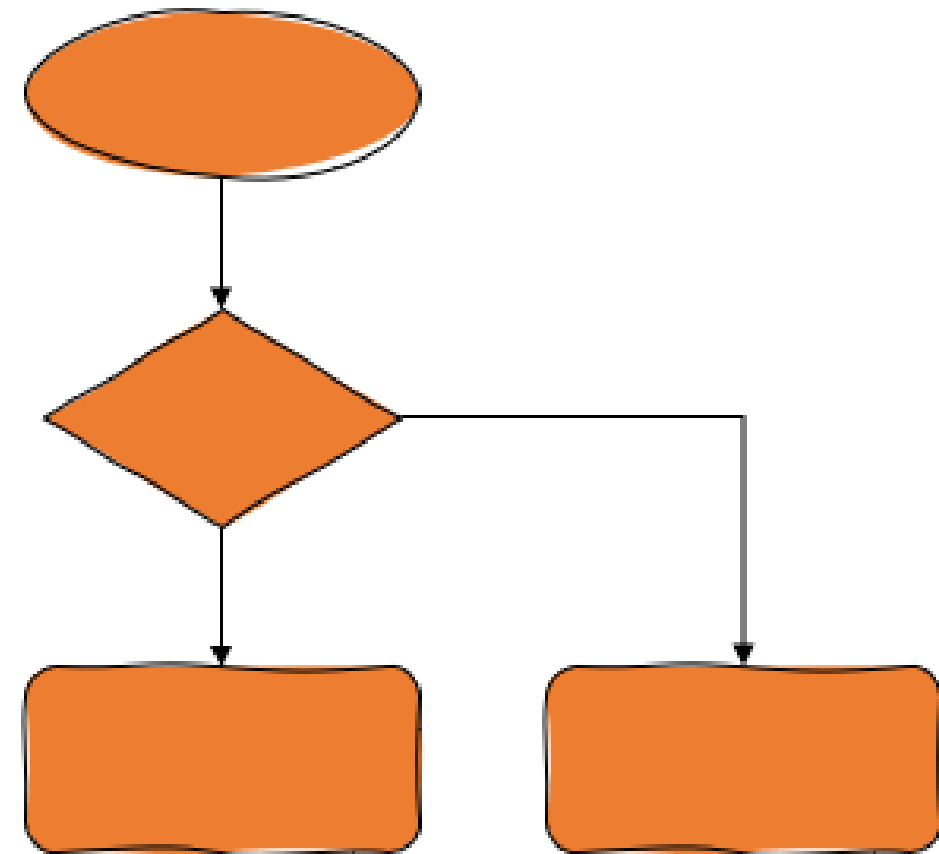


# Coming up!

Model life-cycling



MLOps architecture





# Let's practice!

MLOPS DEPLOYMENT AND LIFE CYCLING

# Life-cycling stages

MLOPS DEPLOYMENT AND LIFE CYCLING



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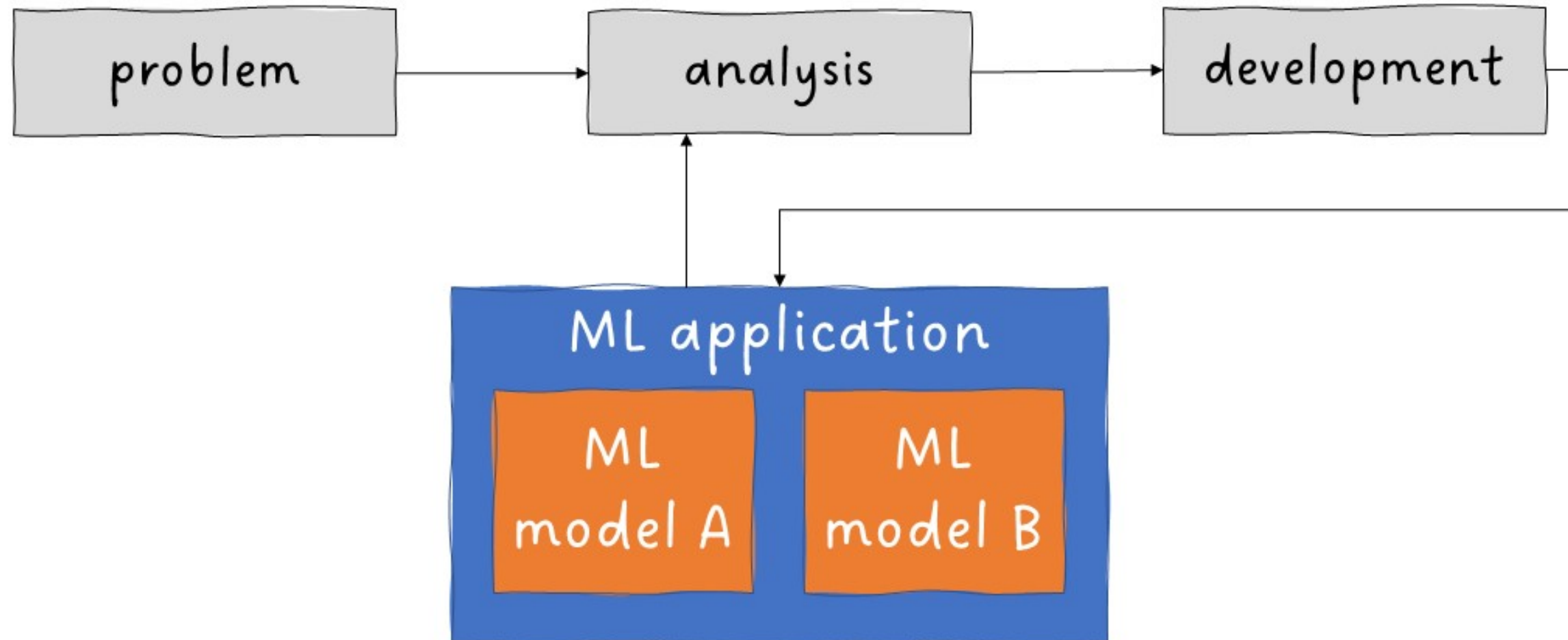
# Machine Learning life cycles

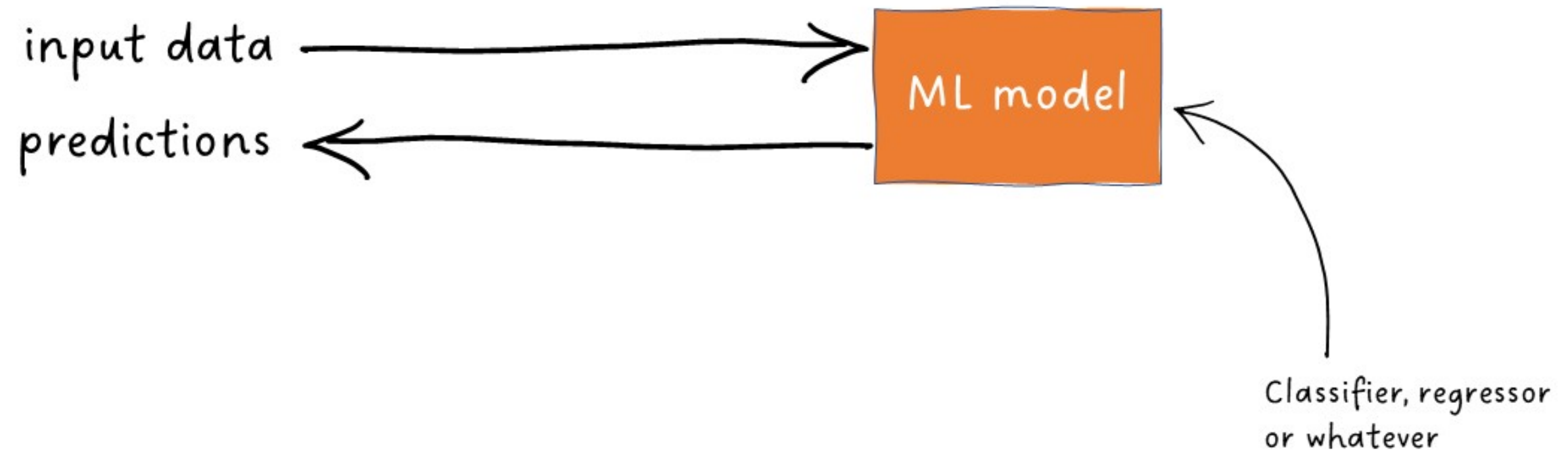
1. ML project life cycle
2. ML application life cycle
3. ML model life cycle

# ML PROJECT

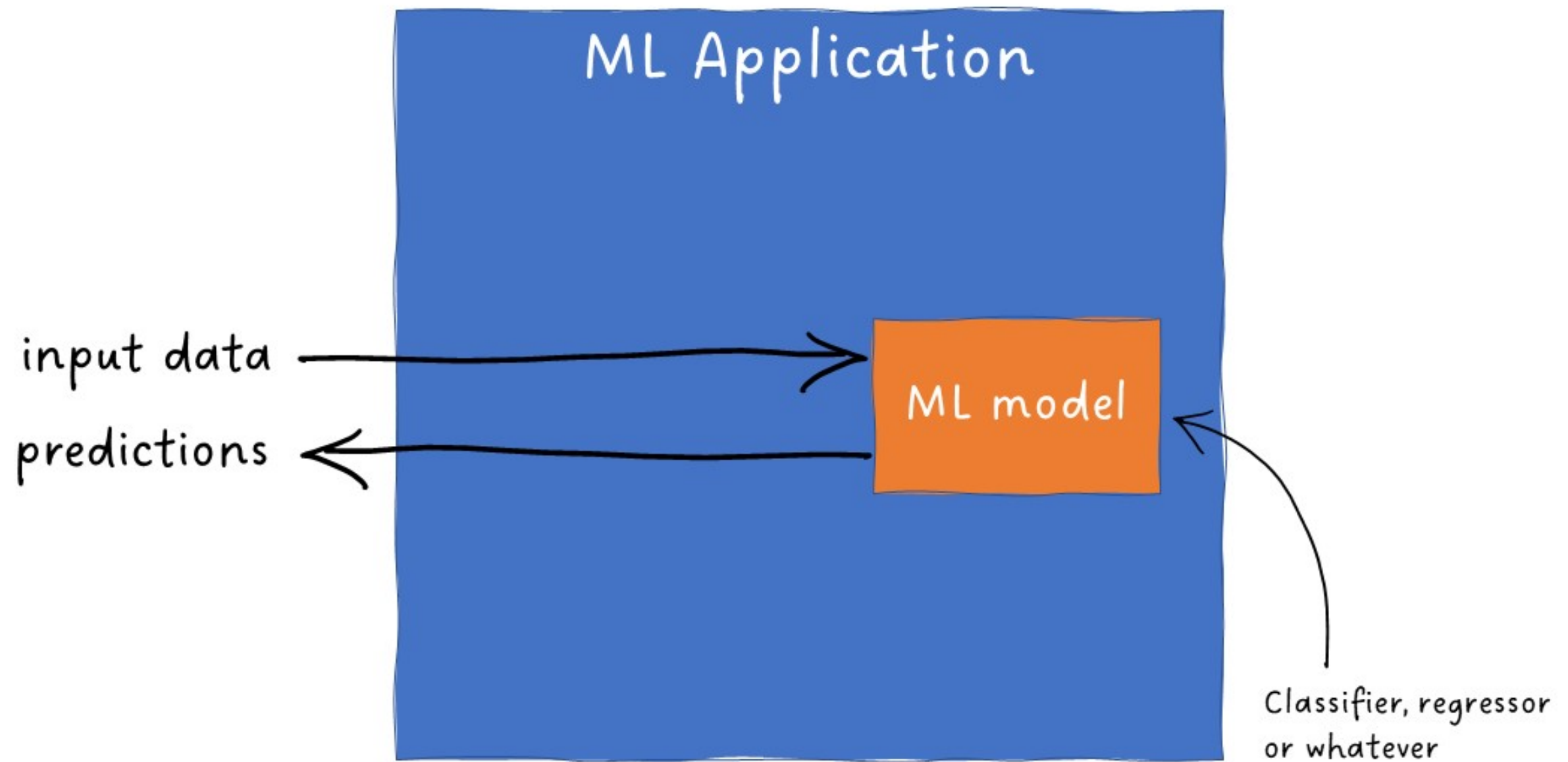


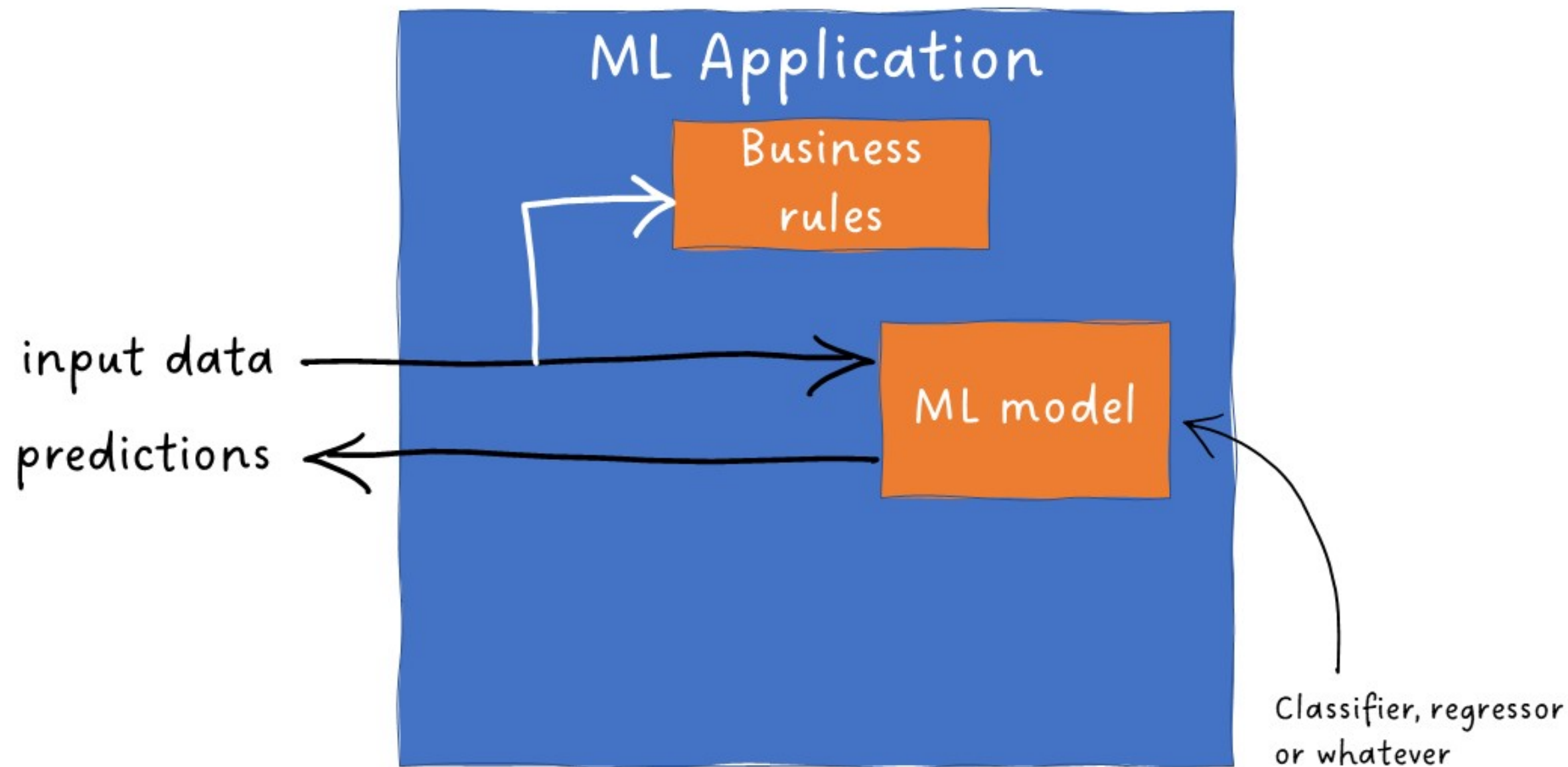
# ML PROJECT

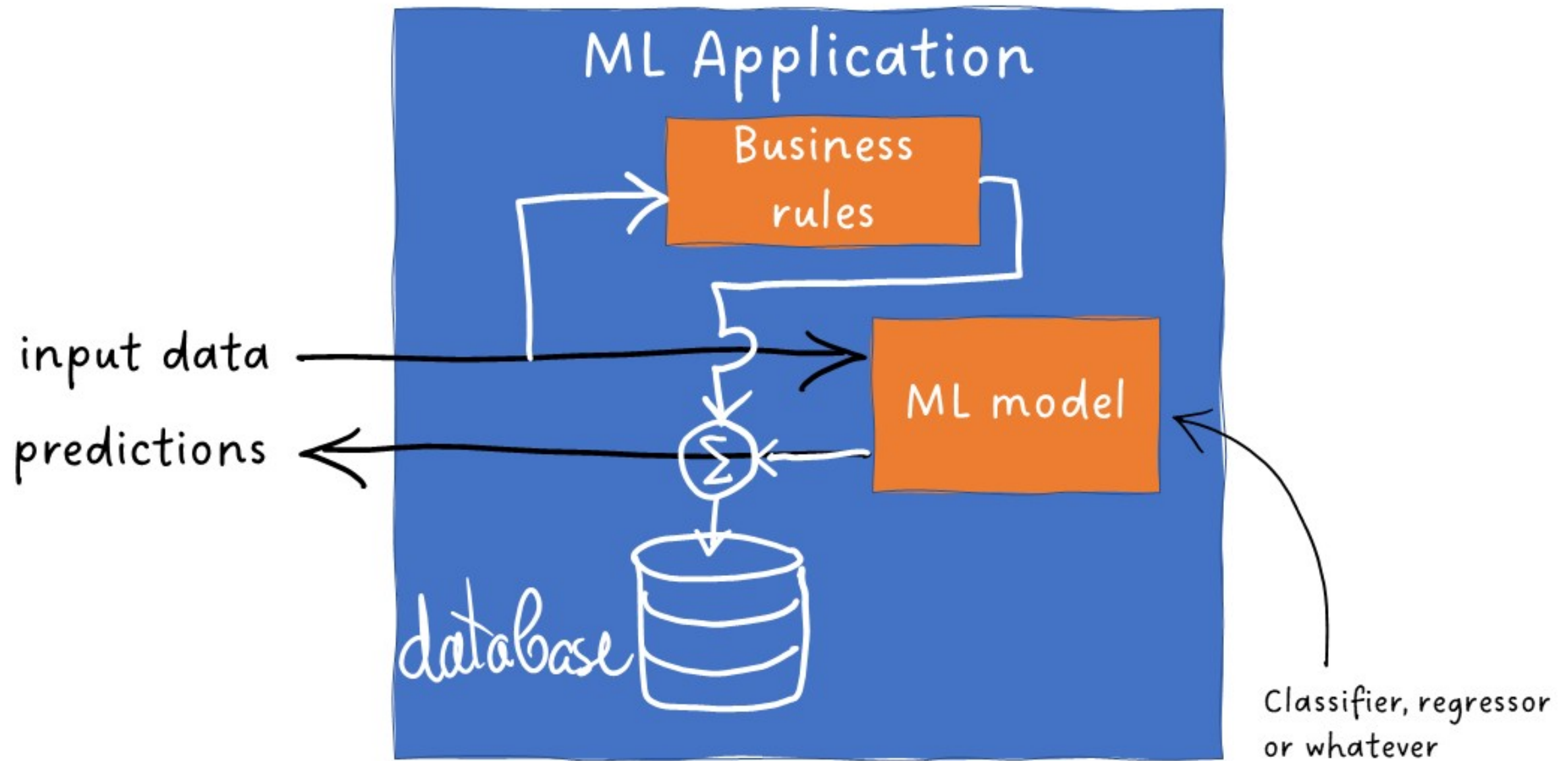


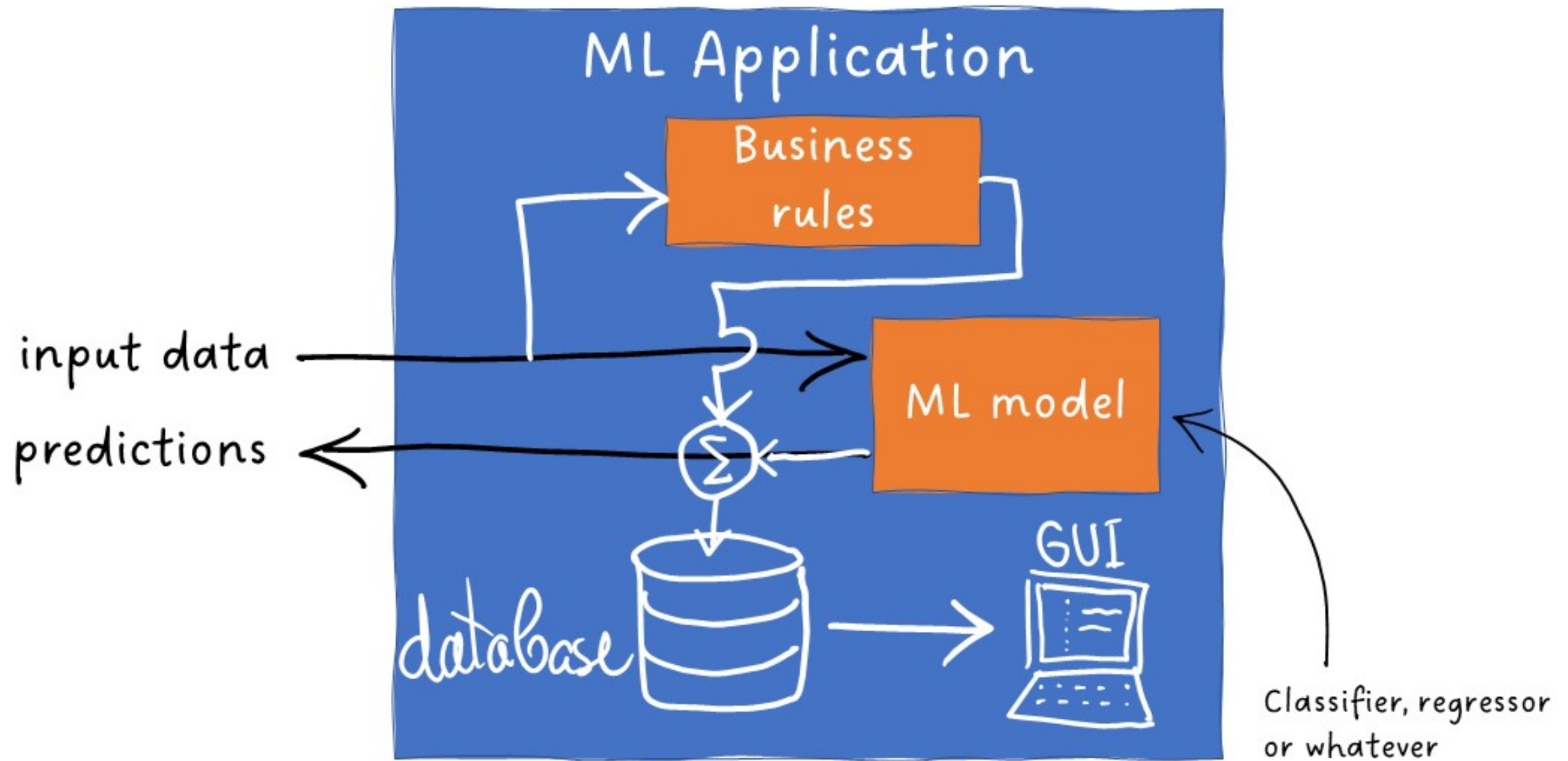


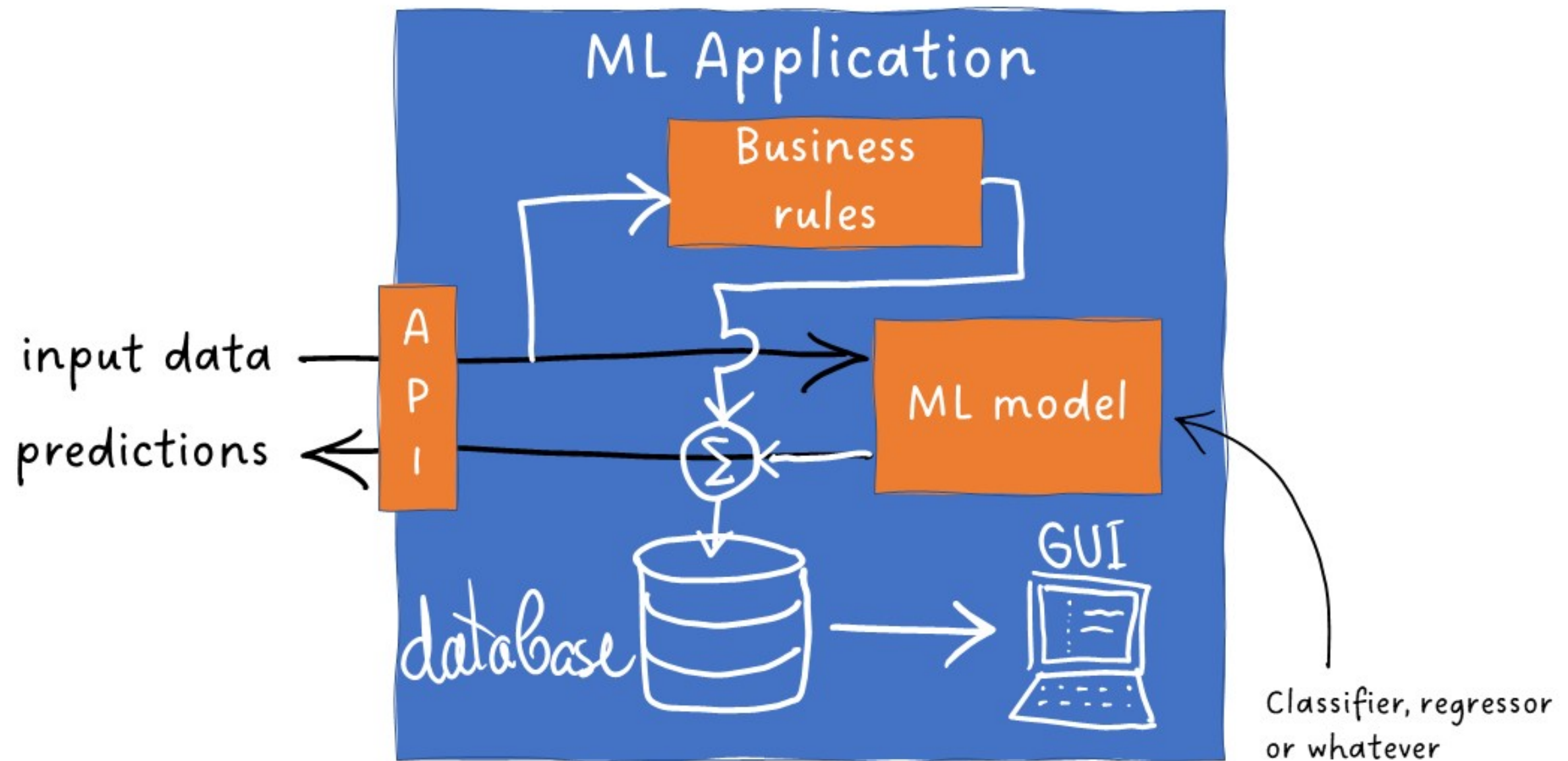






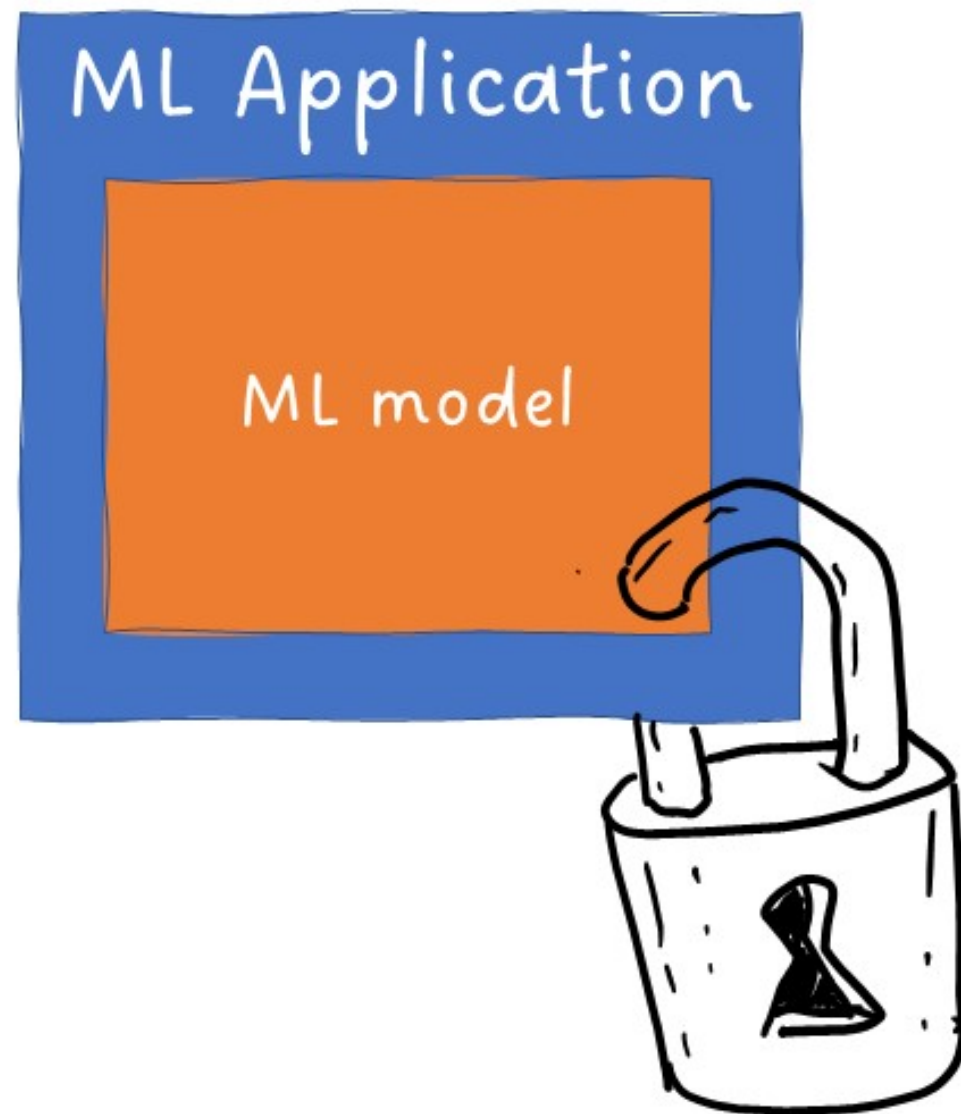




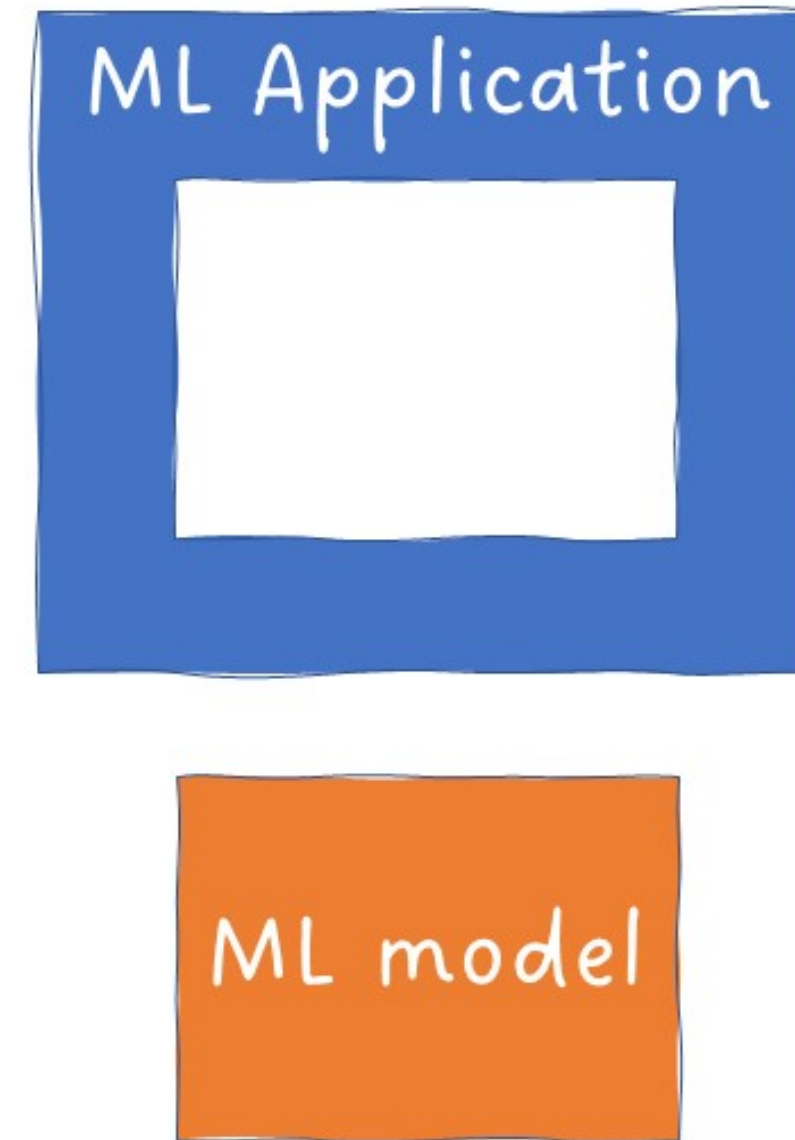
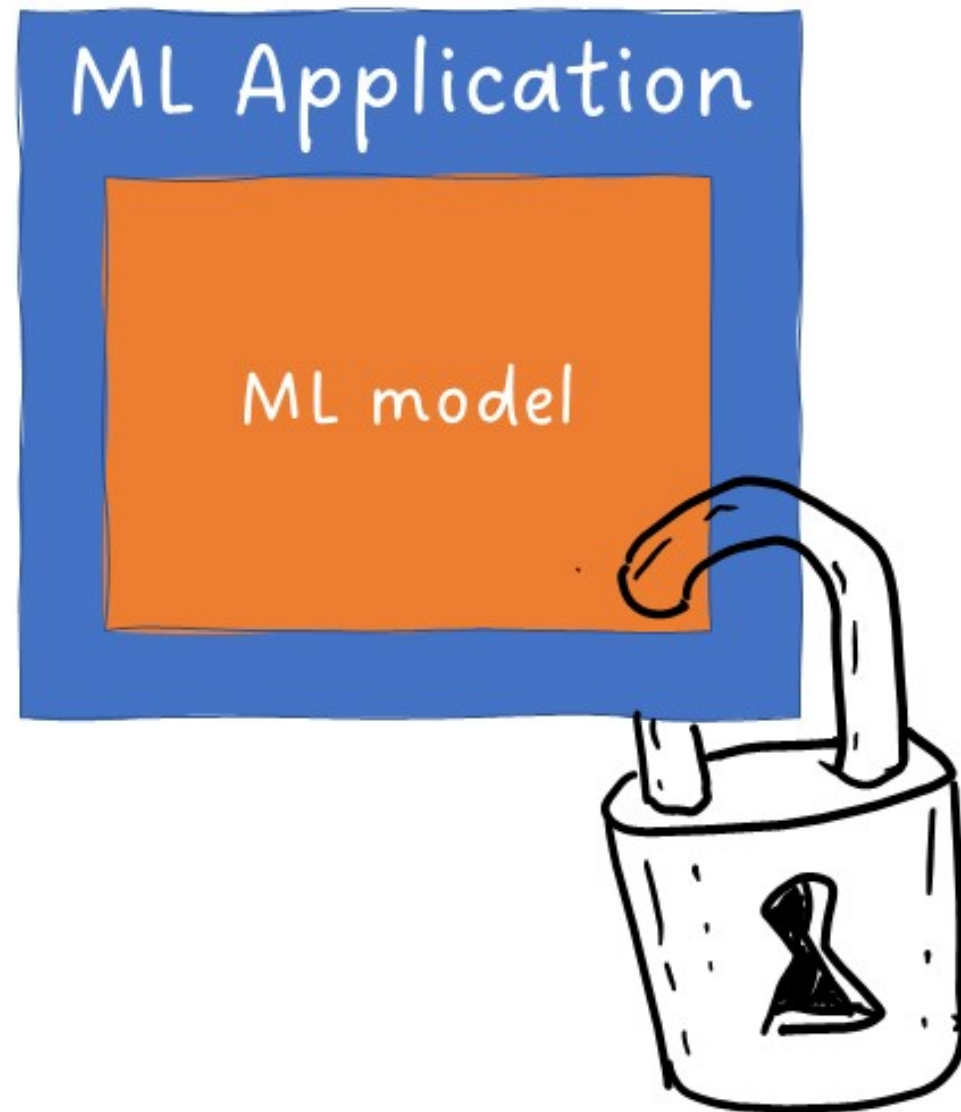


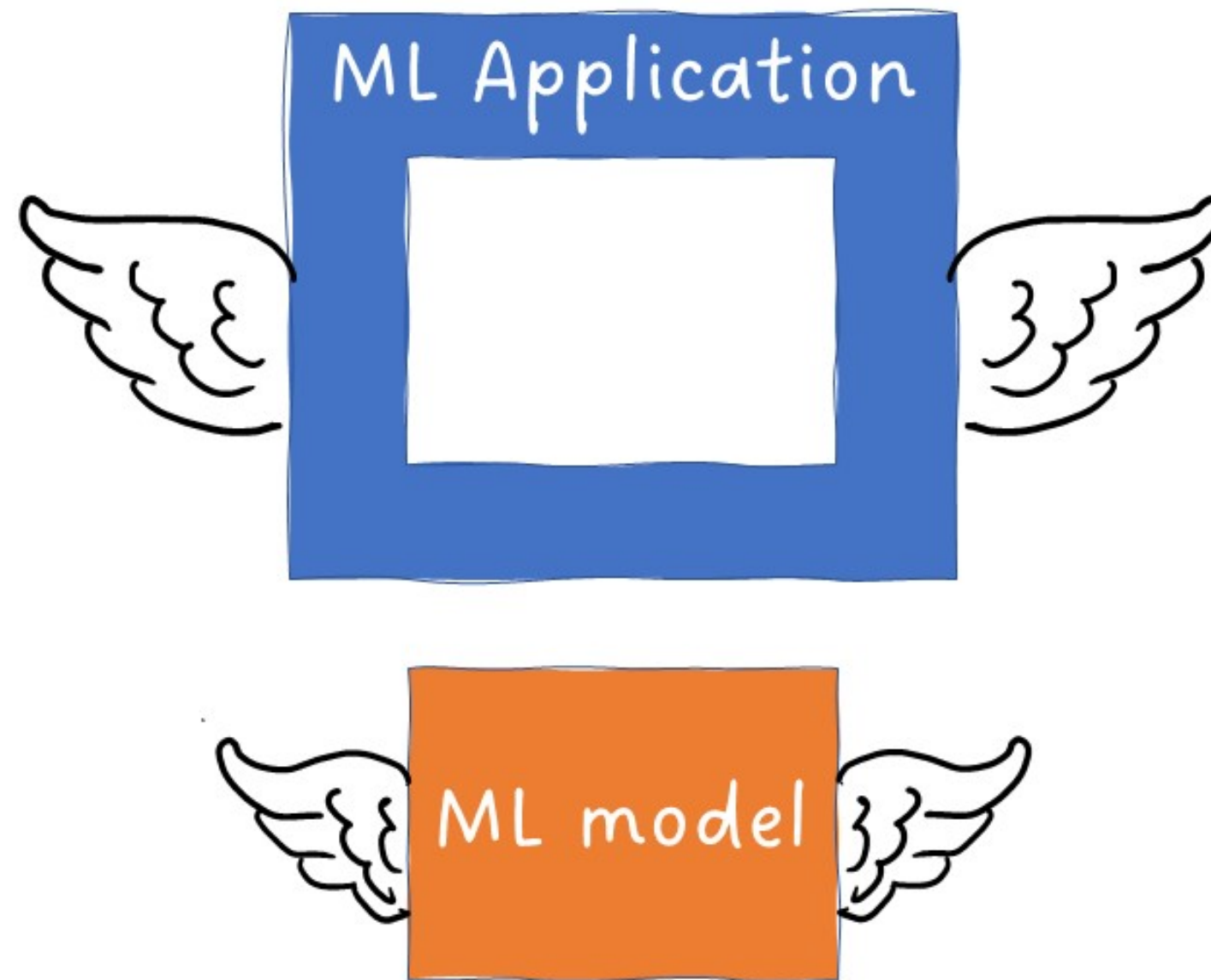
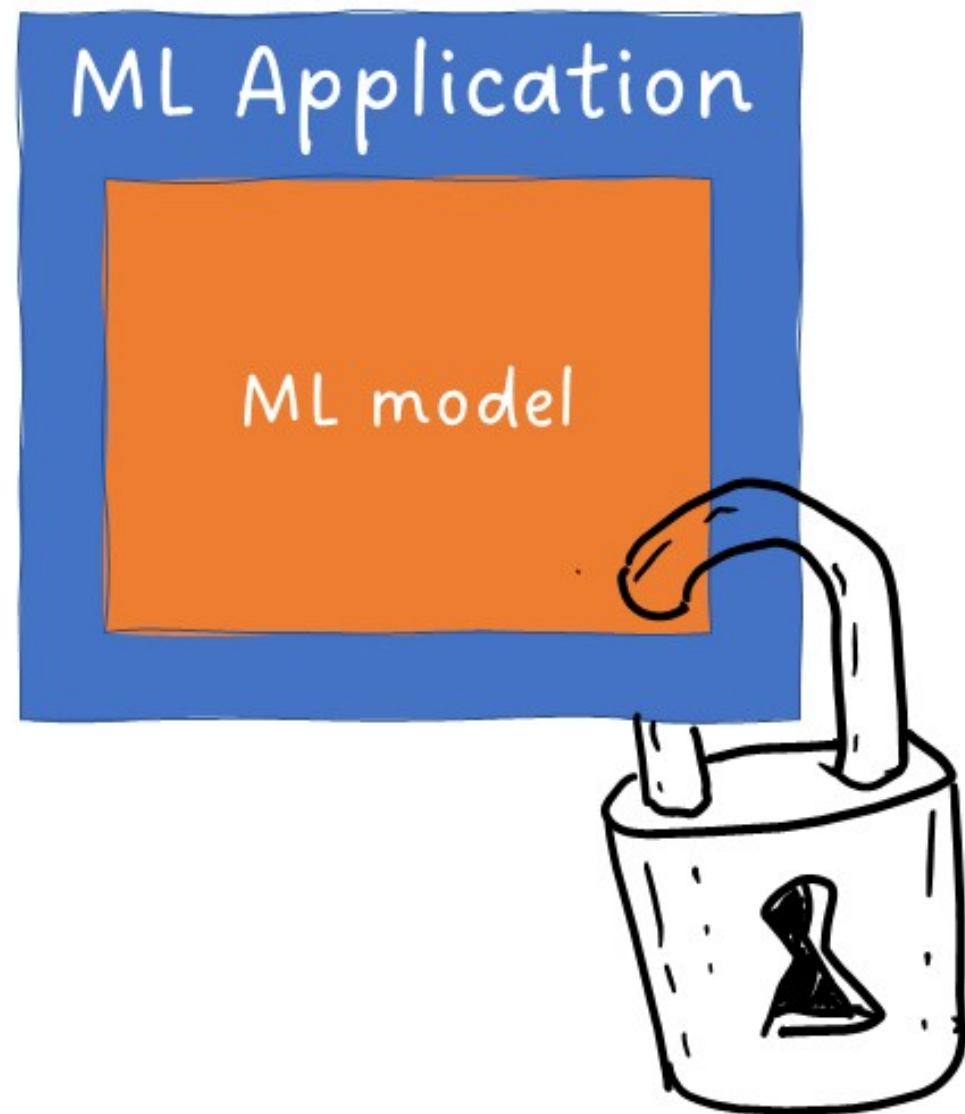




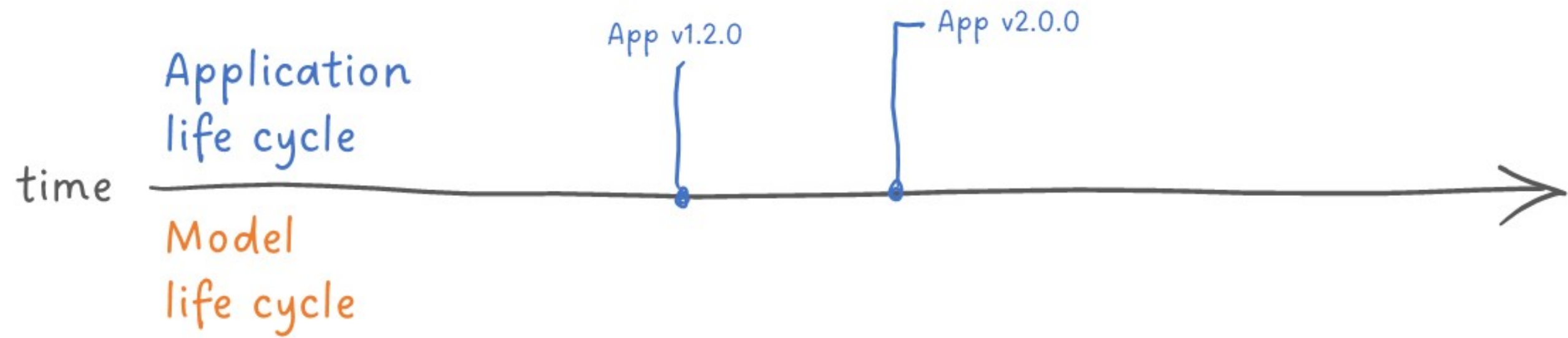


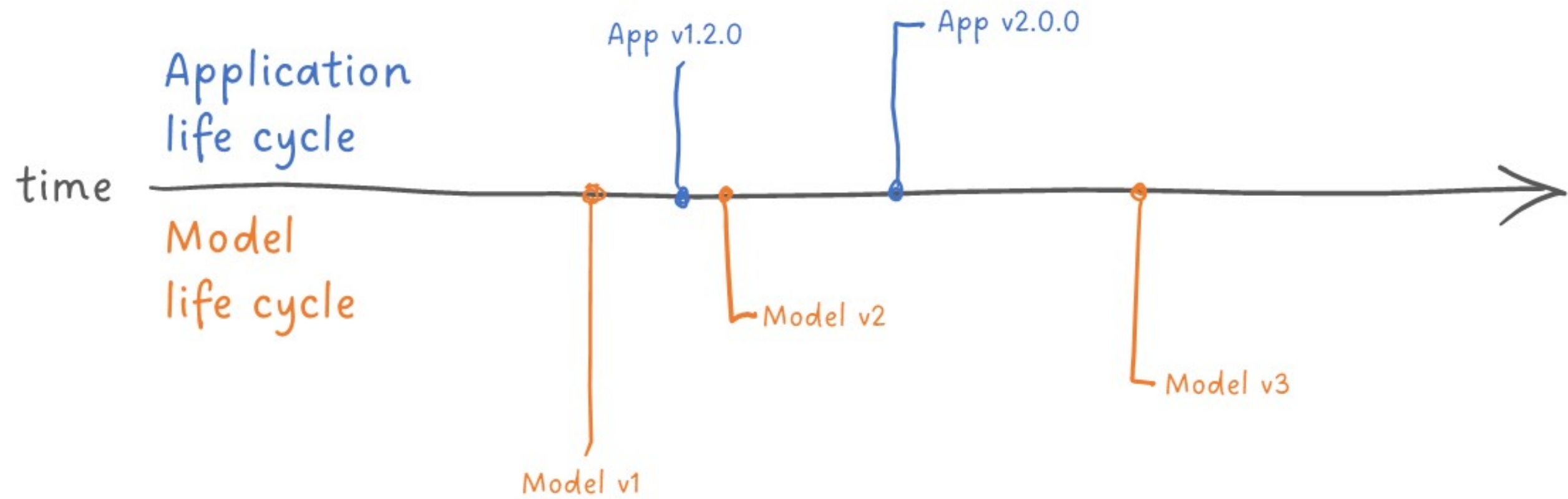


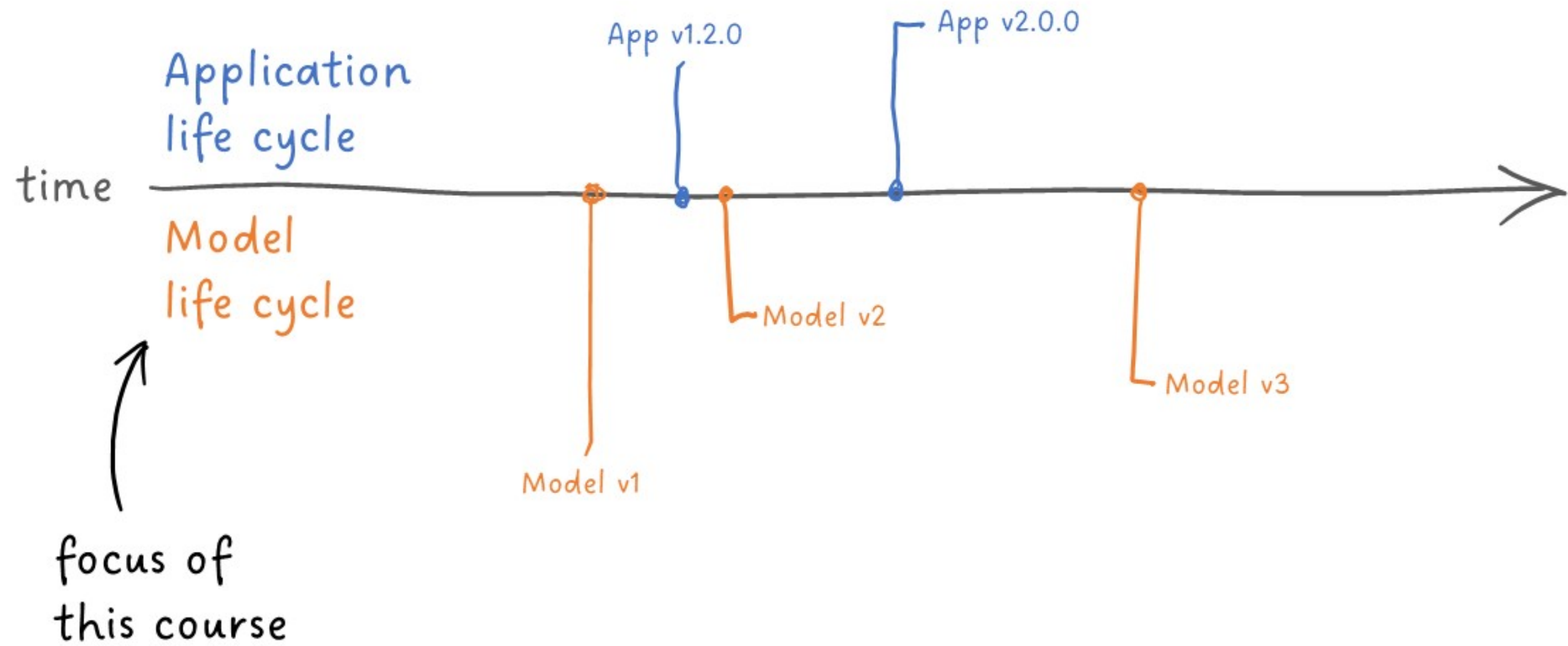












# A model is...

NOT something abstract and/or theoretical

## A model IS

- a concrete, trained model
- ready to be put in use



# Deployment

- Model object + deployment resources = model deployment package
- Putting our model in use = deployment



# Model monitoring

- Is the model server running?
- Are the model inputs and outputs as expected?
- Also known as *post-deployment monitoring*



# Decommissioning



## Possible reasons

- Better model discovered
- More informative features created
- Modeled process changed, invalidating the existing model

=> Out with the old, in with the new



# Model archiving

- Important in regulated industries.
  - Why did your credit-risk model green-light this loan back in 2020?
- Reproducibility: ability to recreate, re-load, and use an old version of a model
  - Not a trivial requirement



# Let's practice!

MLOPS DEPLOYMENT AND LIFE CYCLING

# MLOps components

MLOPS DEPLOYMENT AND LIFE CYCLING



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# Concepts

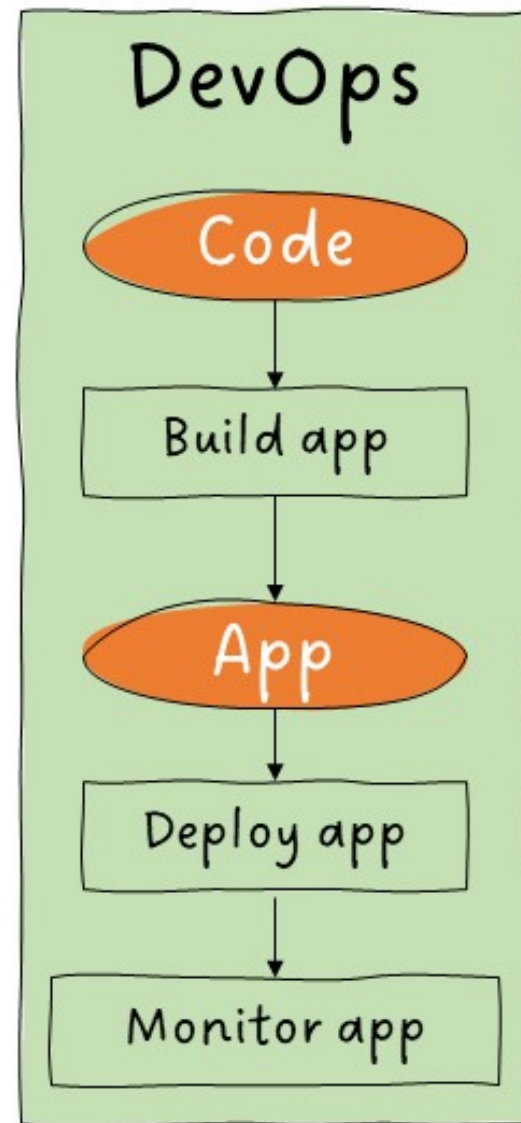
## GENERIC (Software Development)

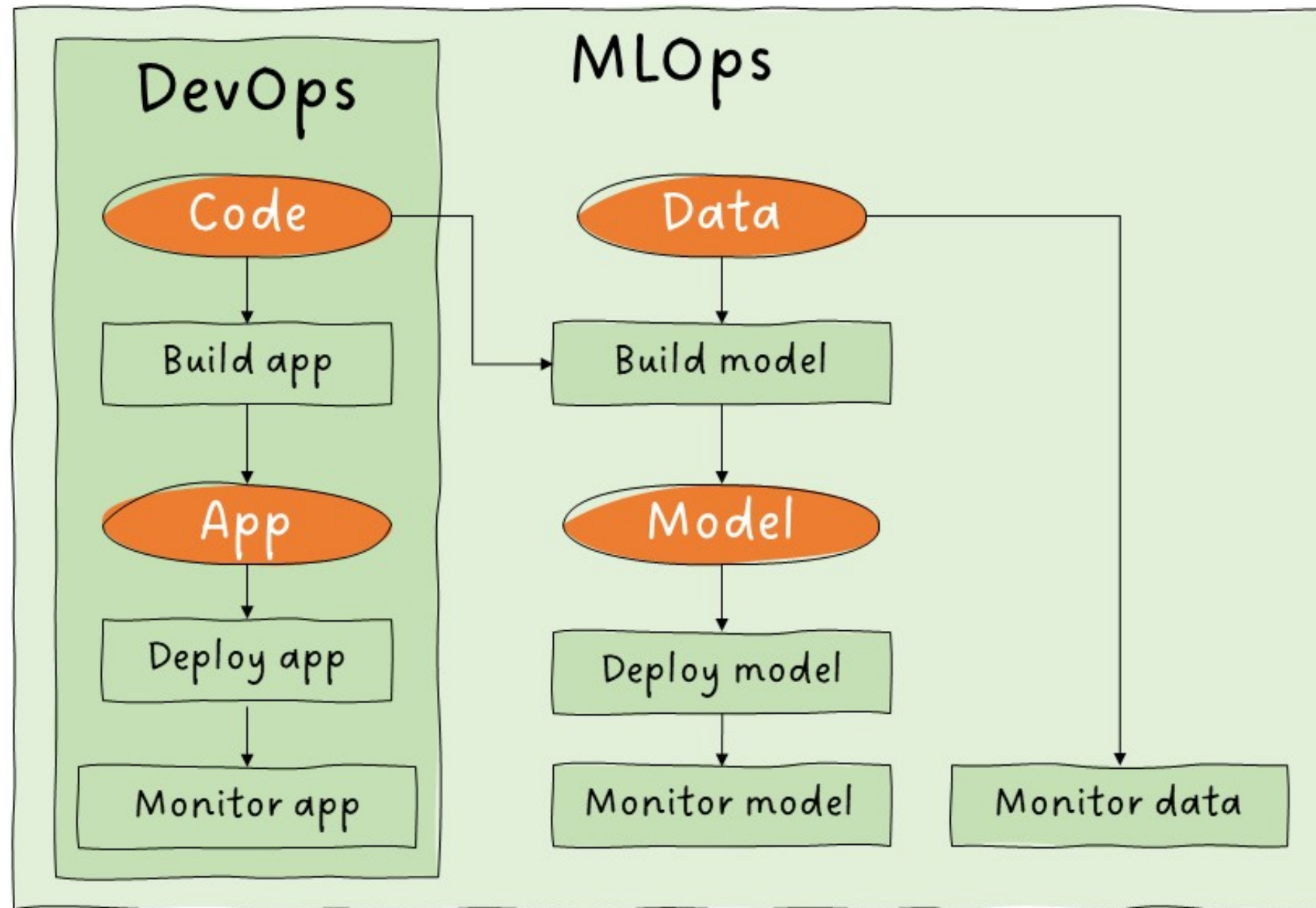
- Workflows
- Pipelines
- Artifacts

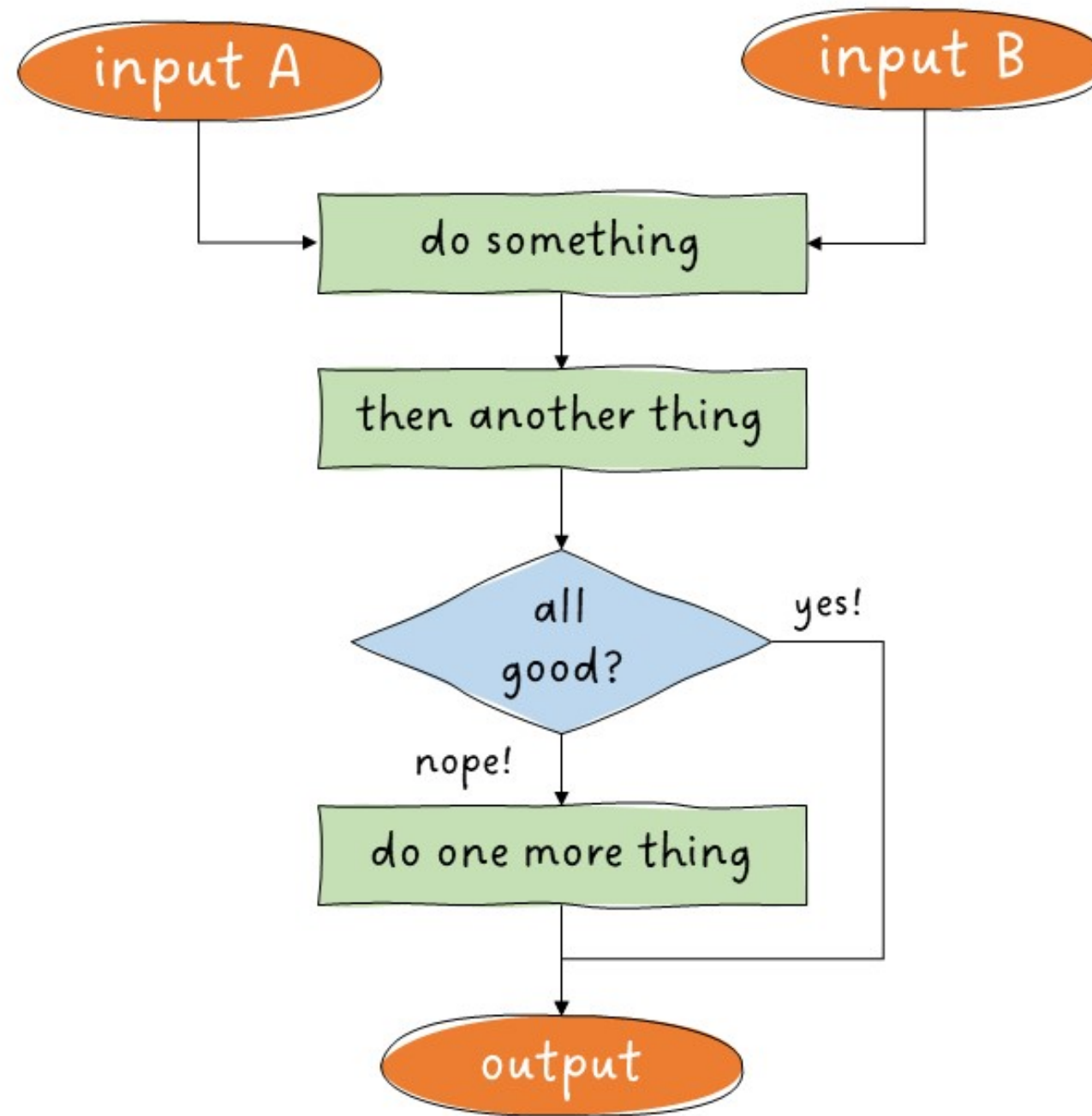
## ML-SPECIFIC

- Model registry
- Feature store
- Metadata store

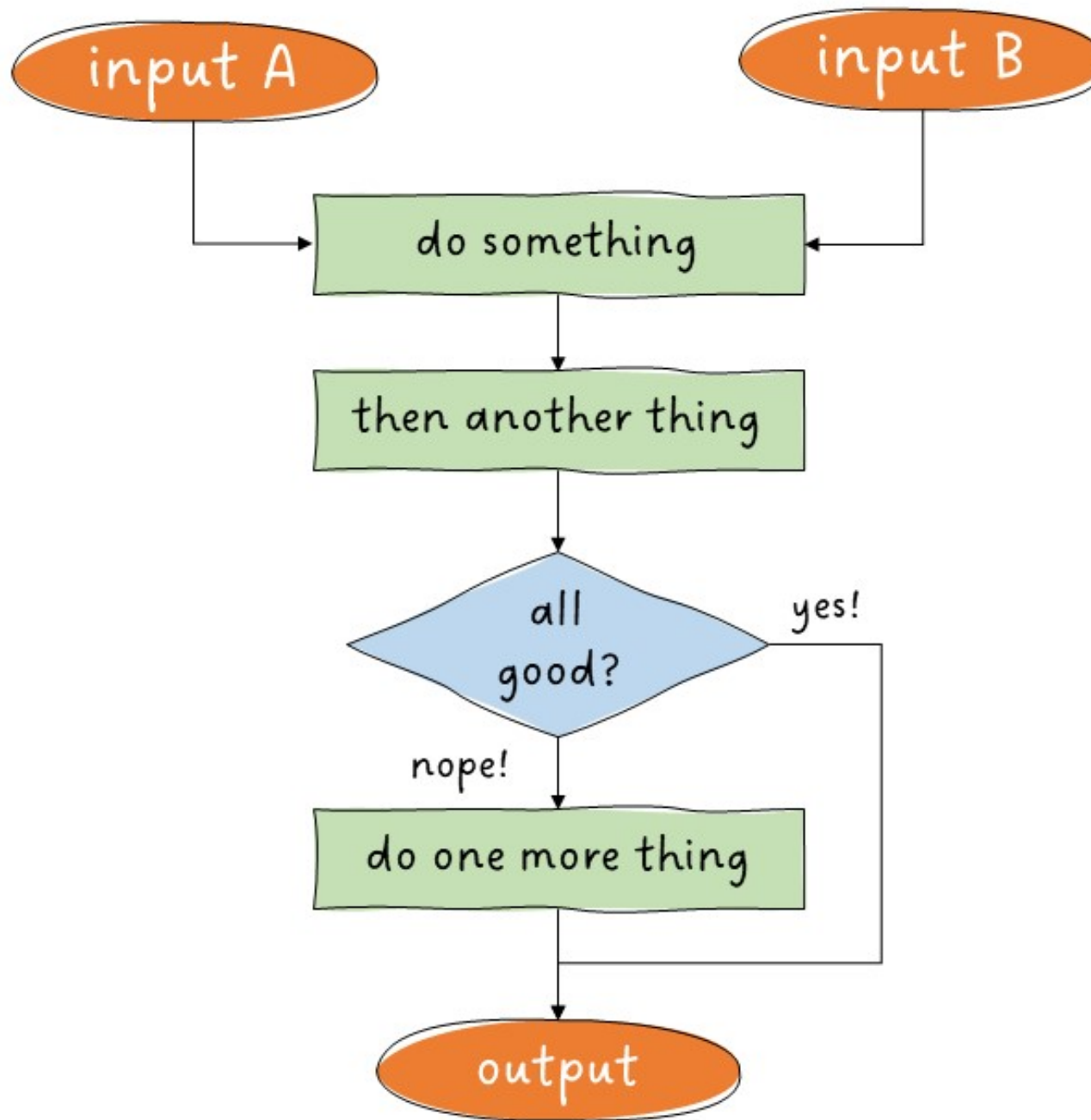






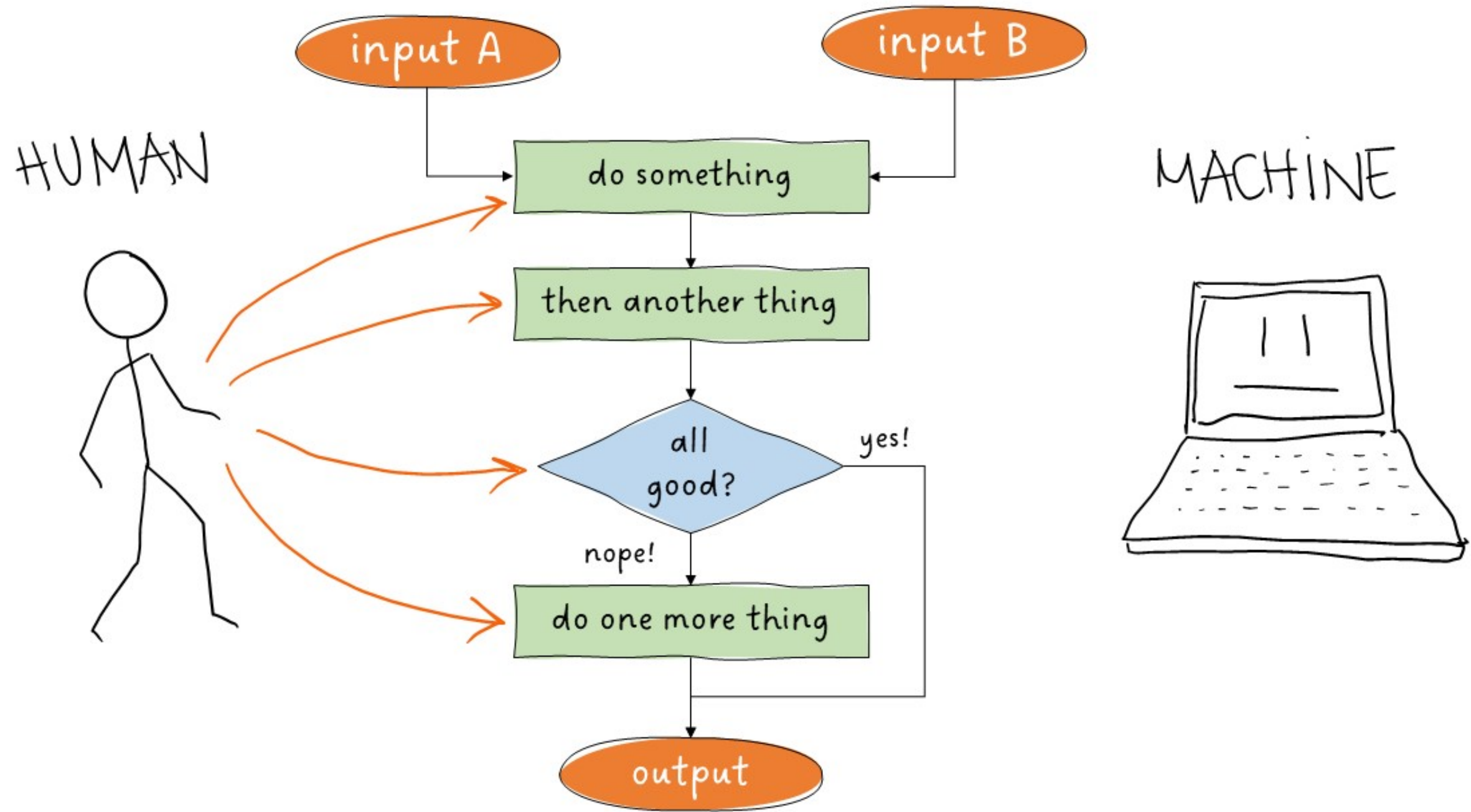


HUMAN

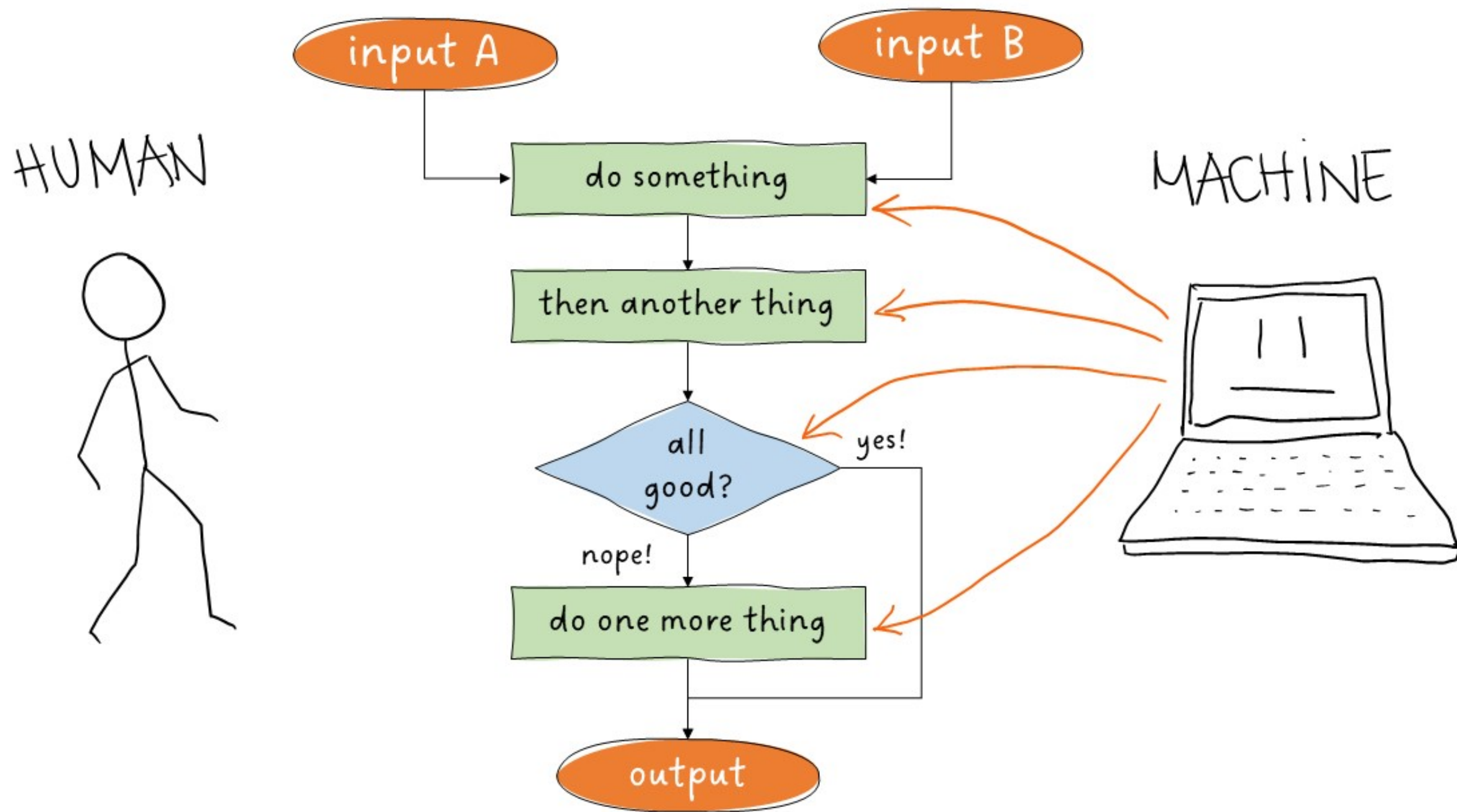


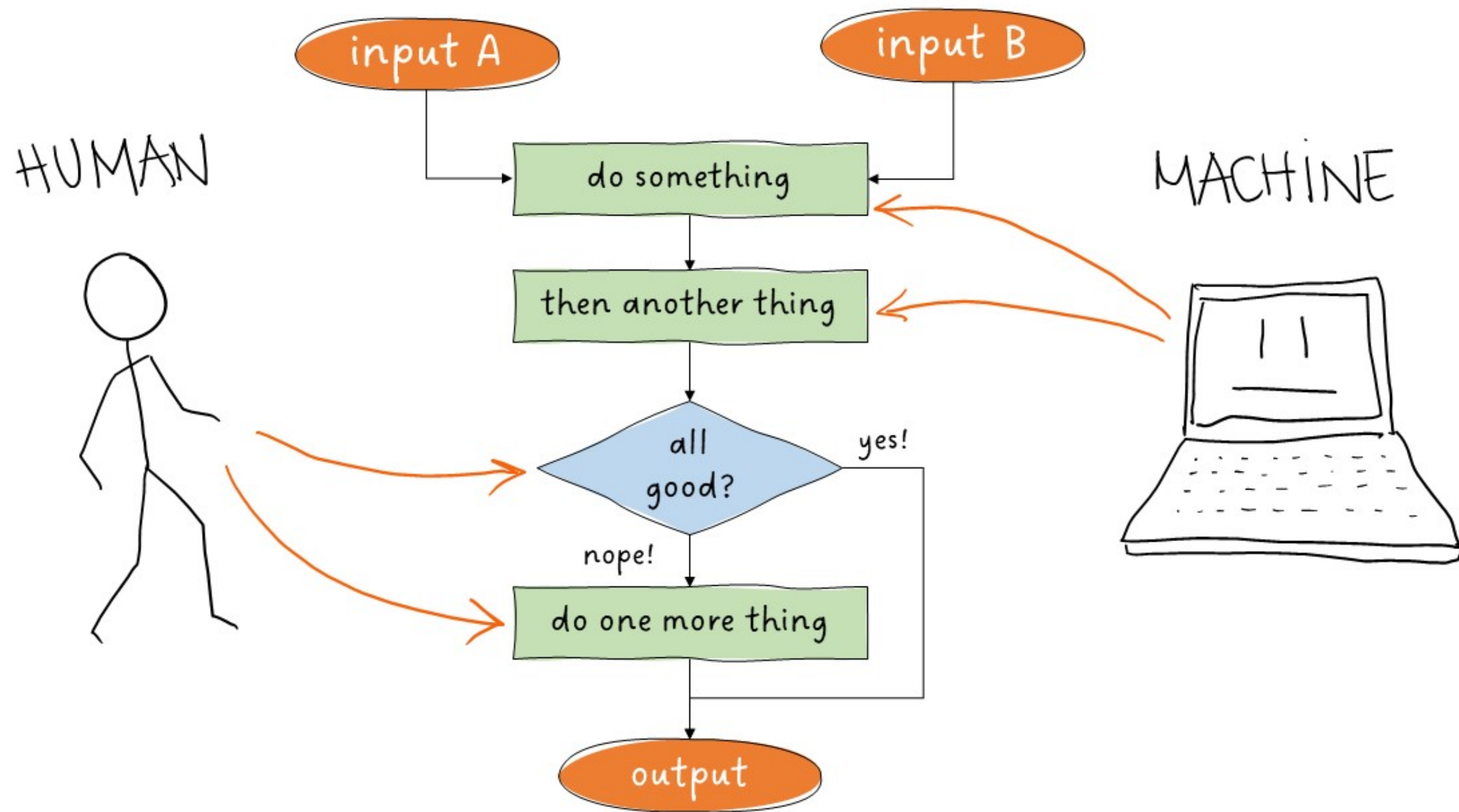
MACHINE



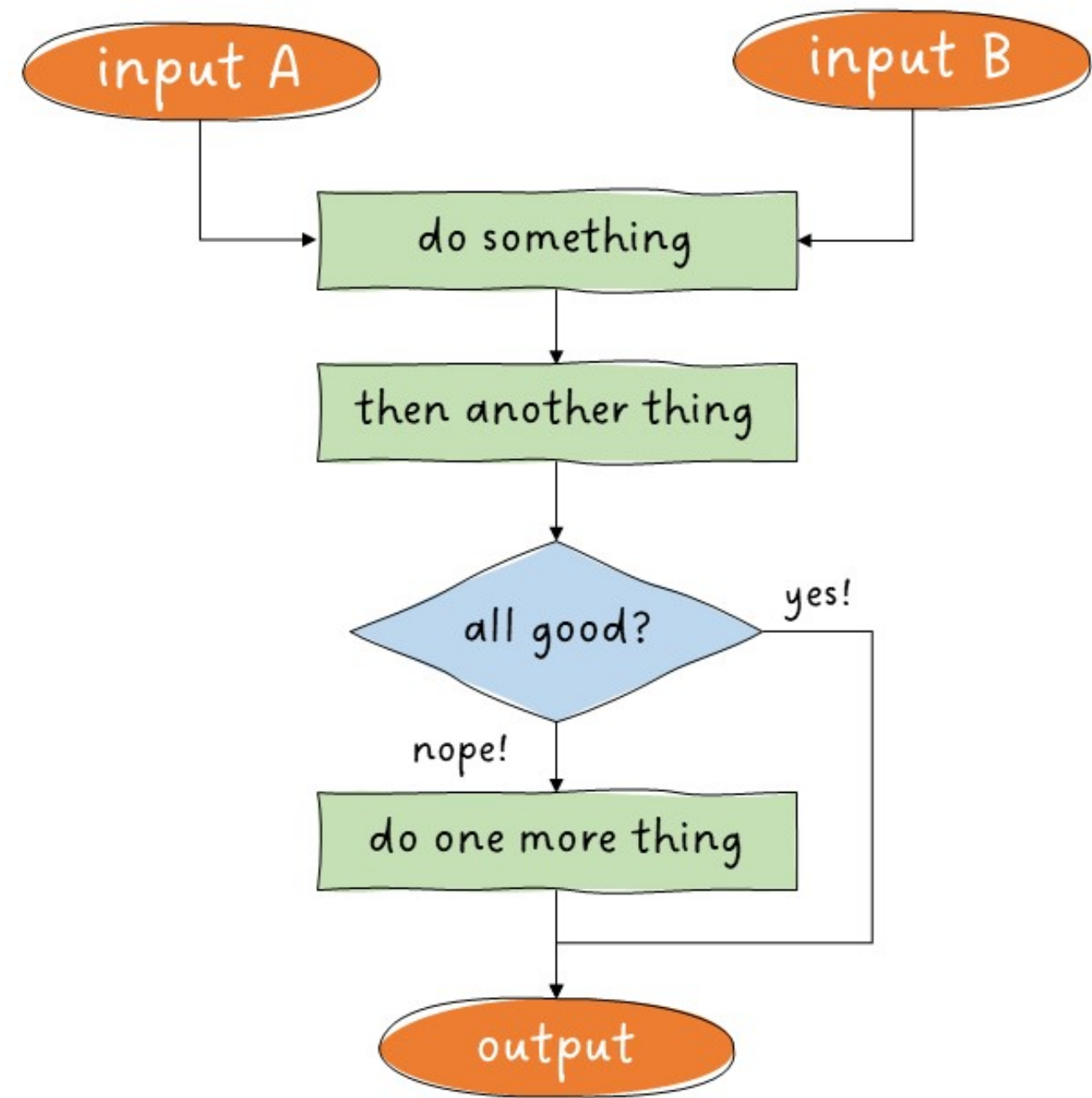


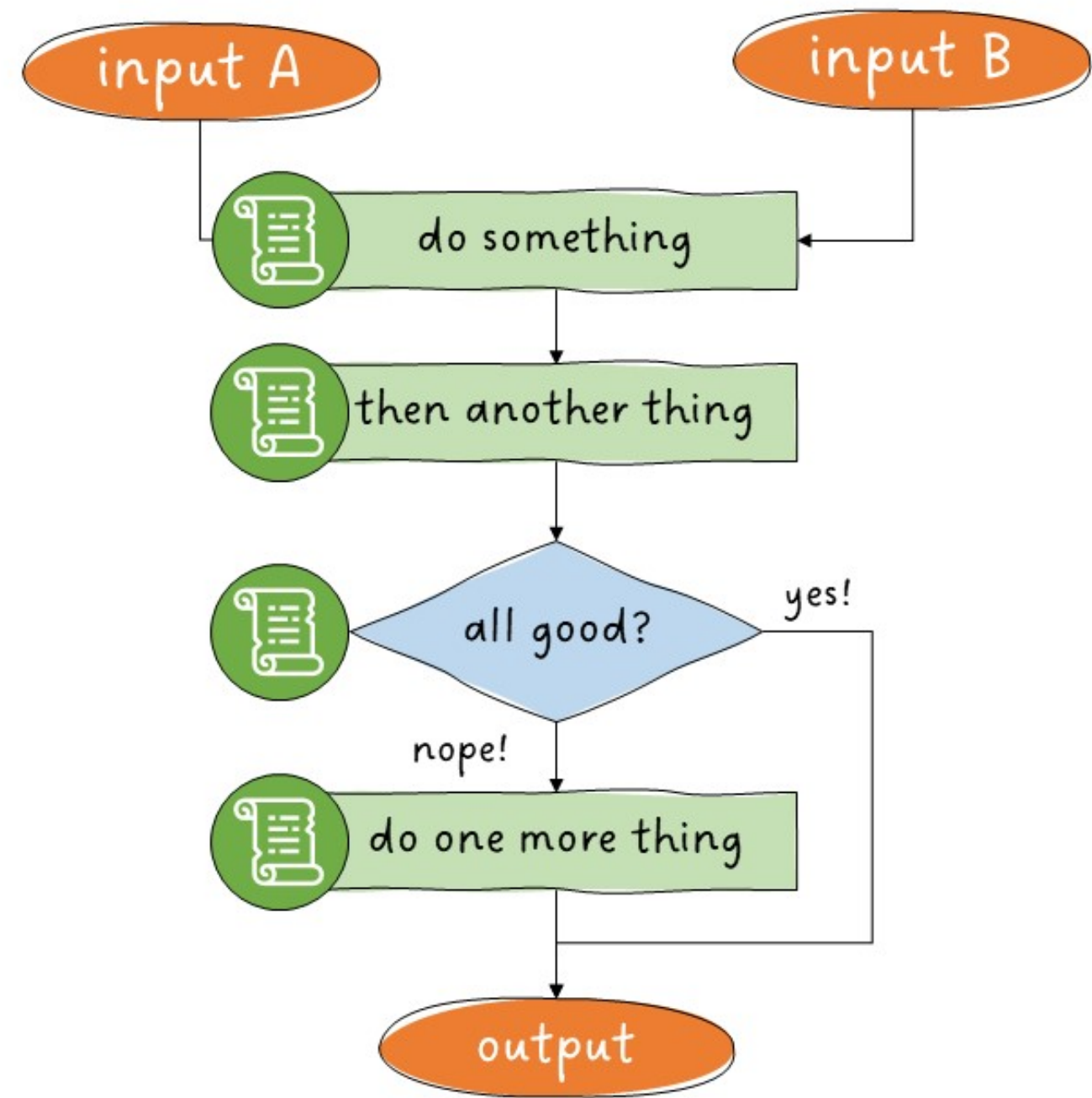


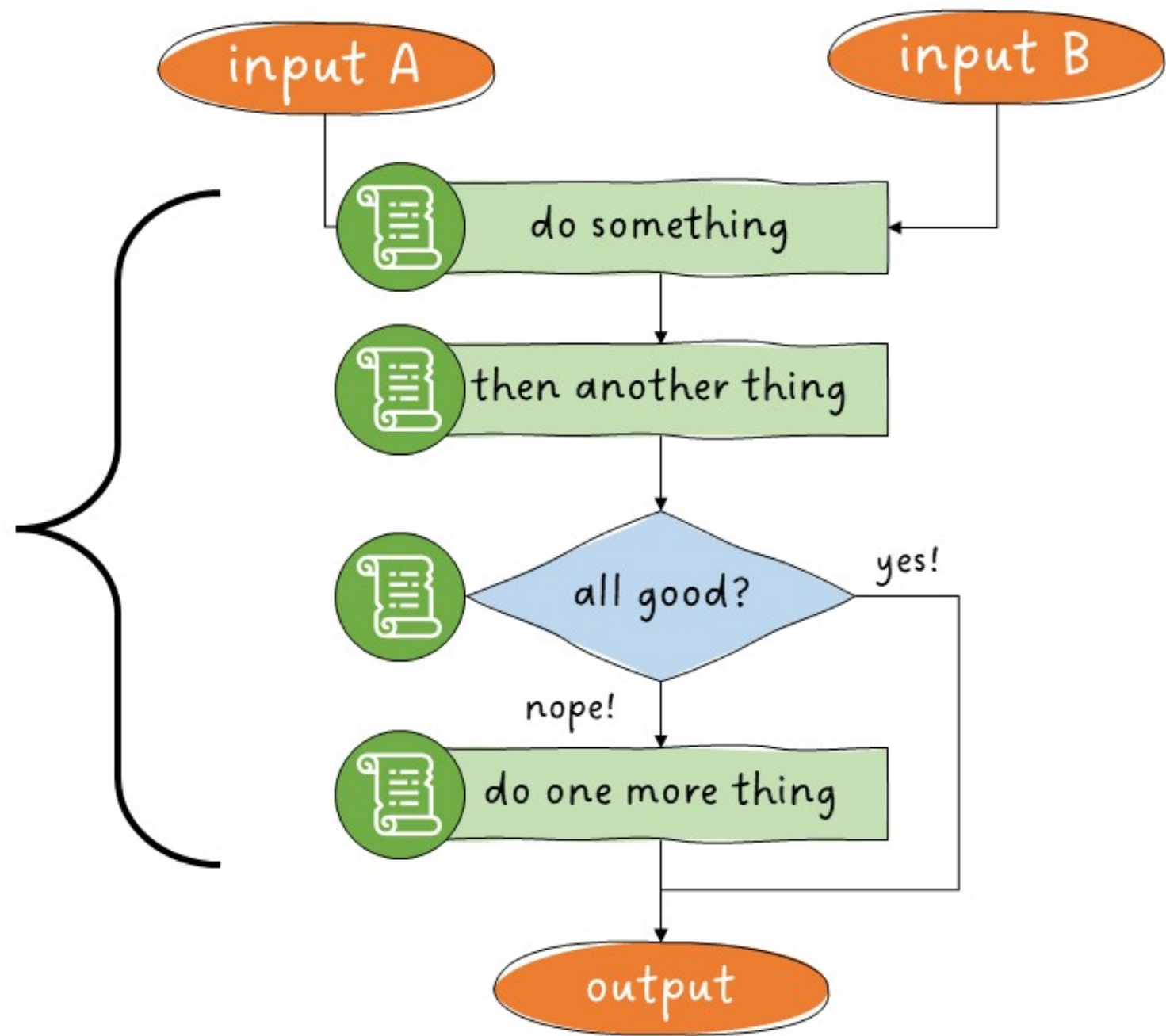
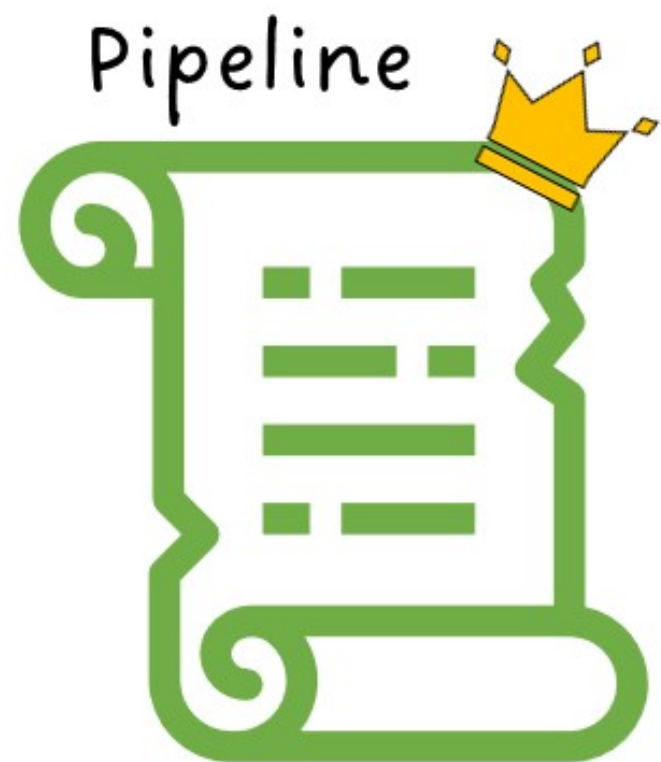


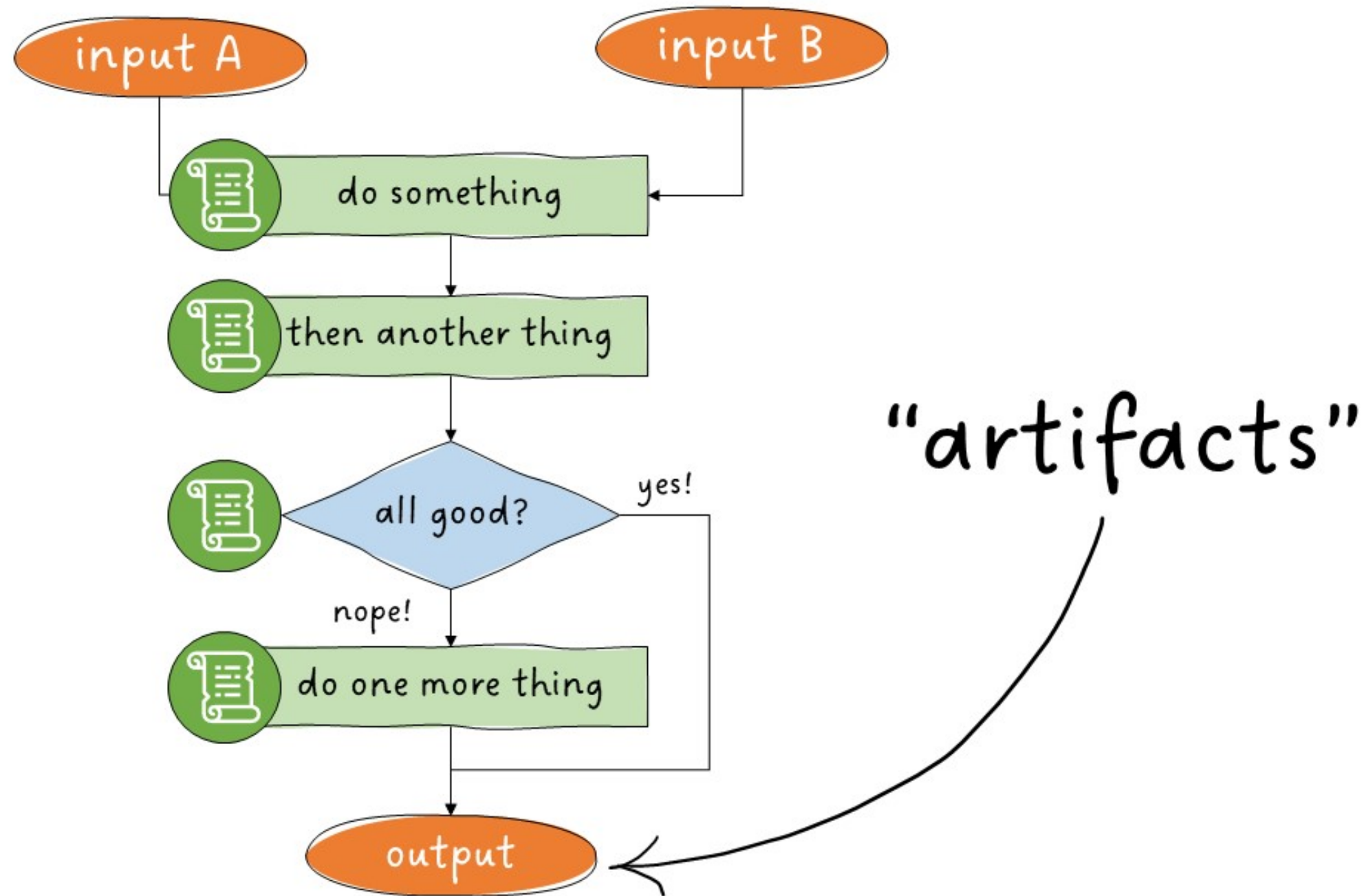




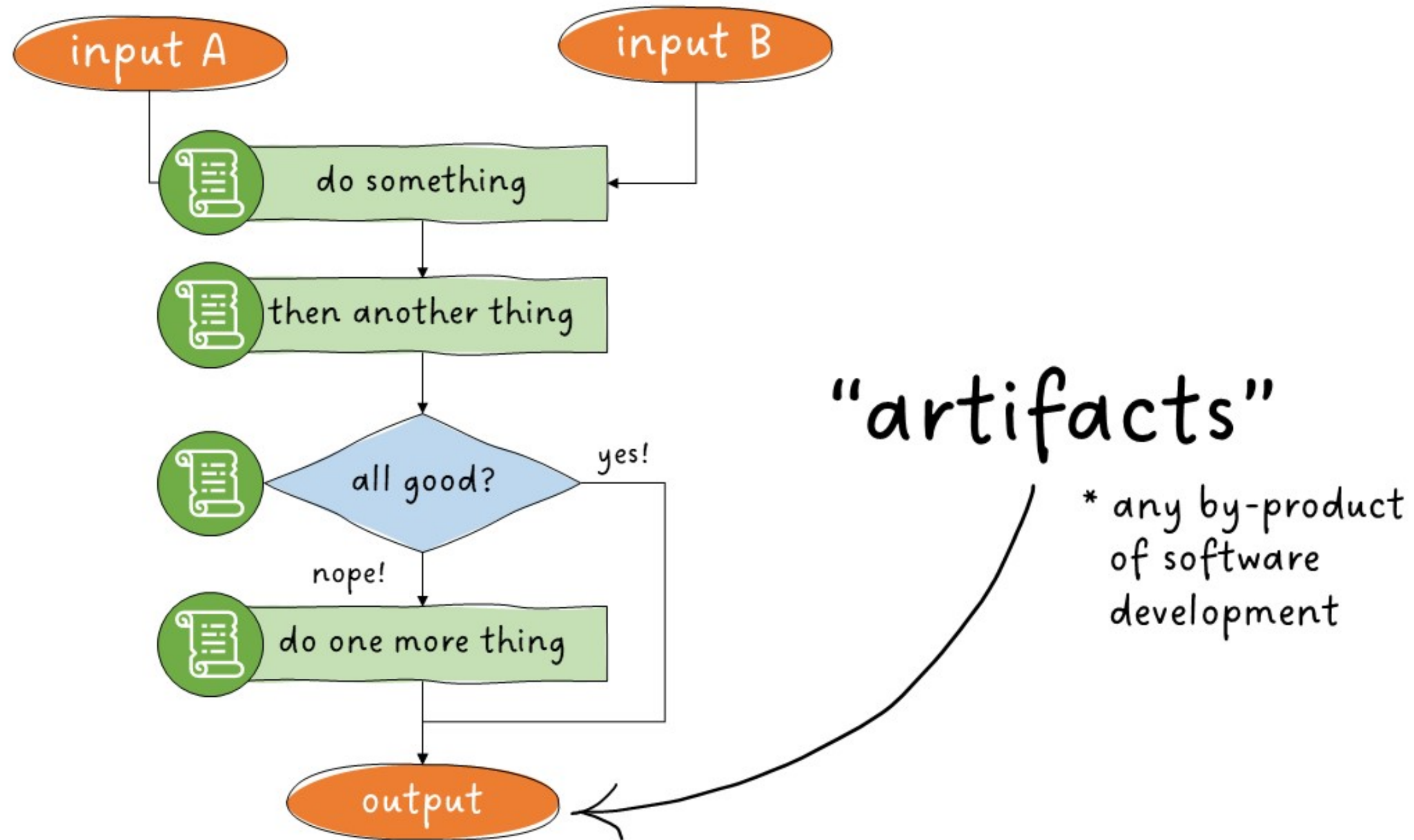








<sup>1</sup> [https://en.wikipedia.org/wiki/Artifact\\_%28software\\_development%29](https://en.wikipedia.org/wiki/Artifact_%28software_development%29)



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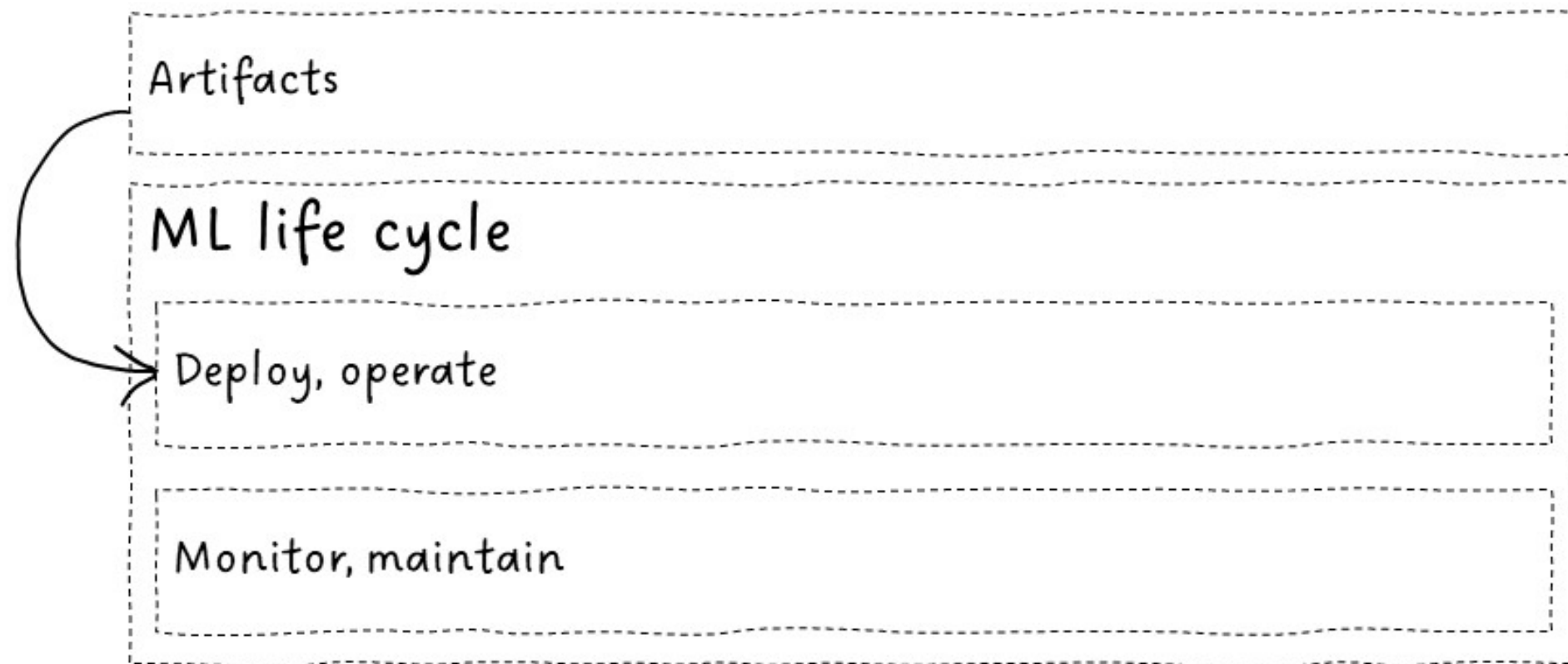


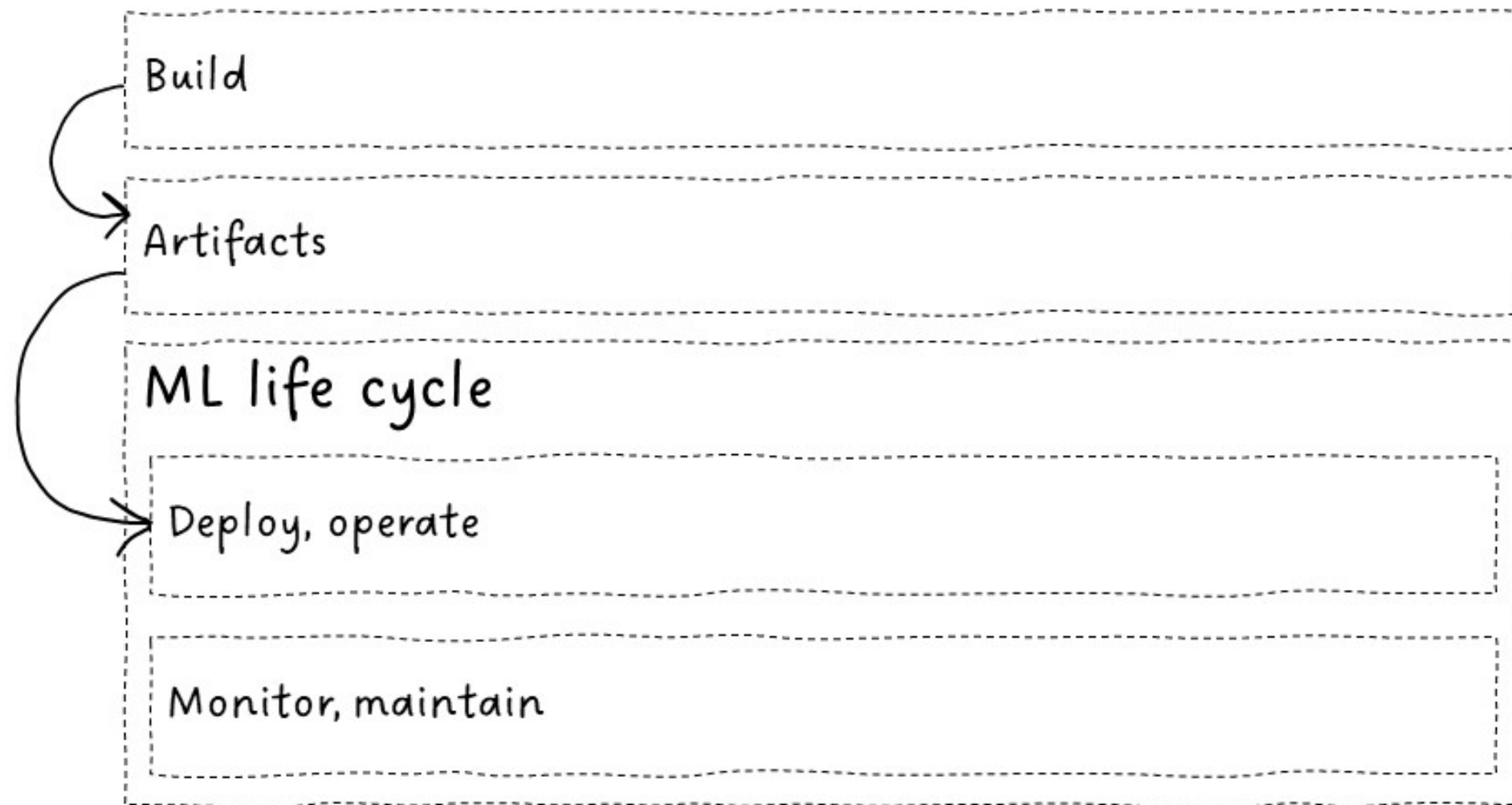
ML life cycle

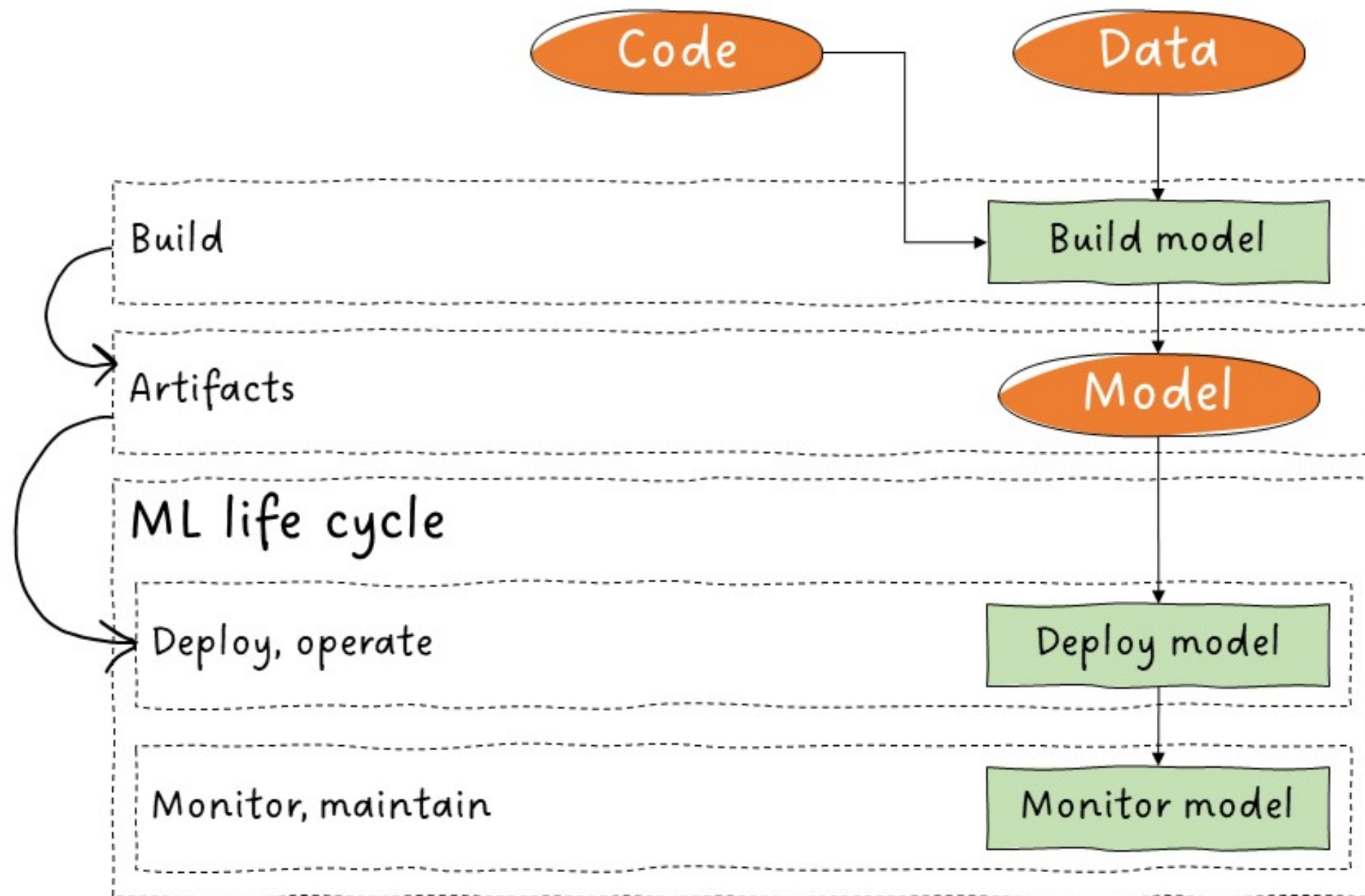
Deploy, operate

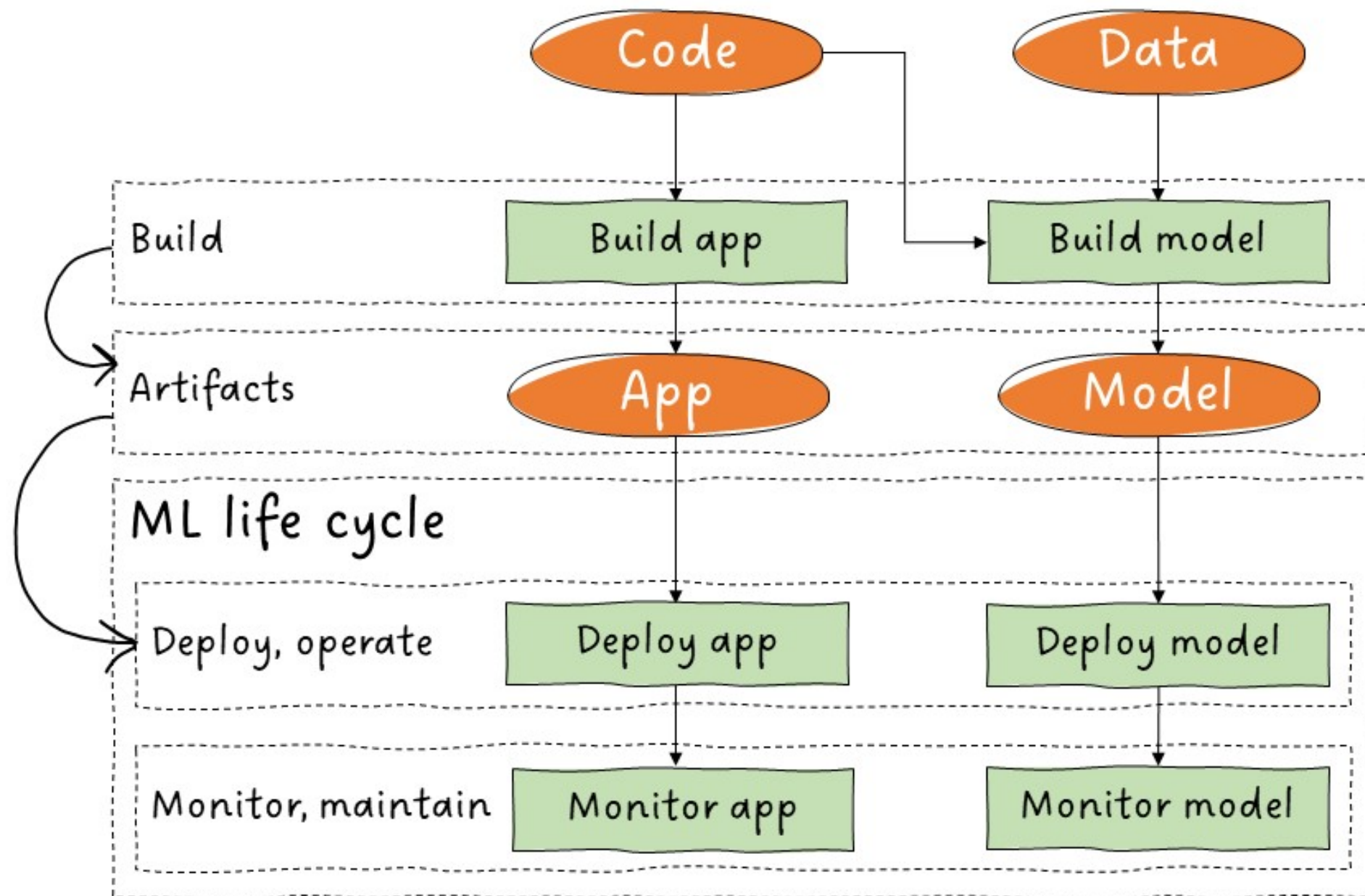
Monitor, maintain



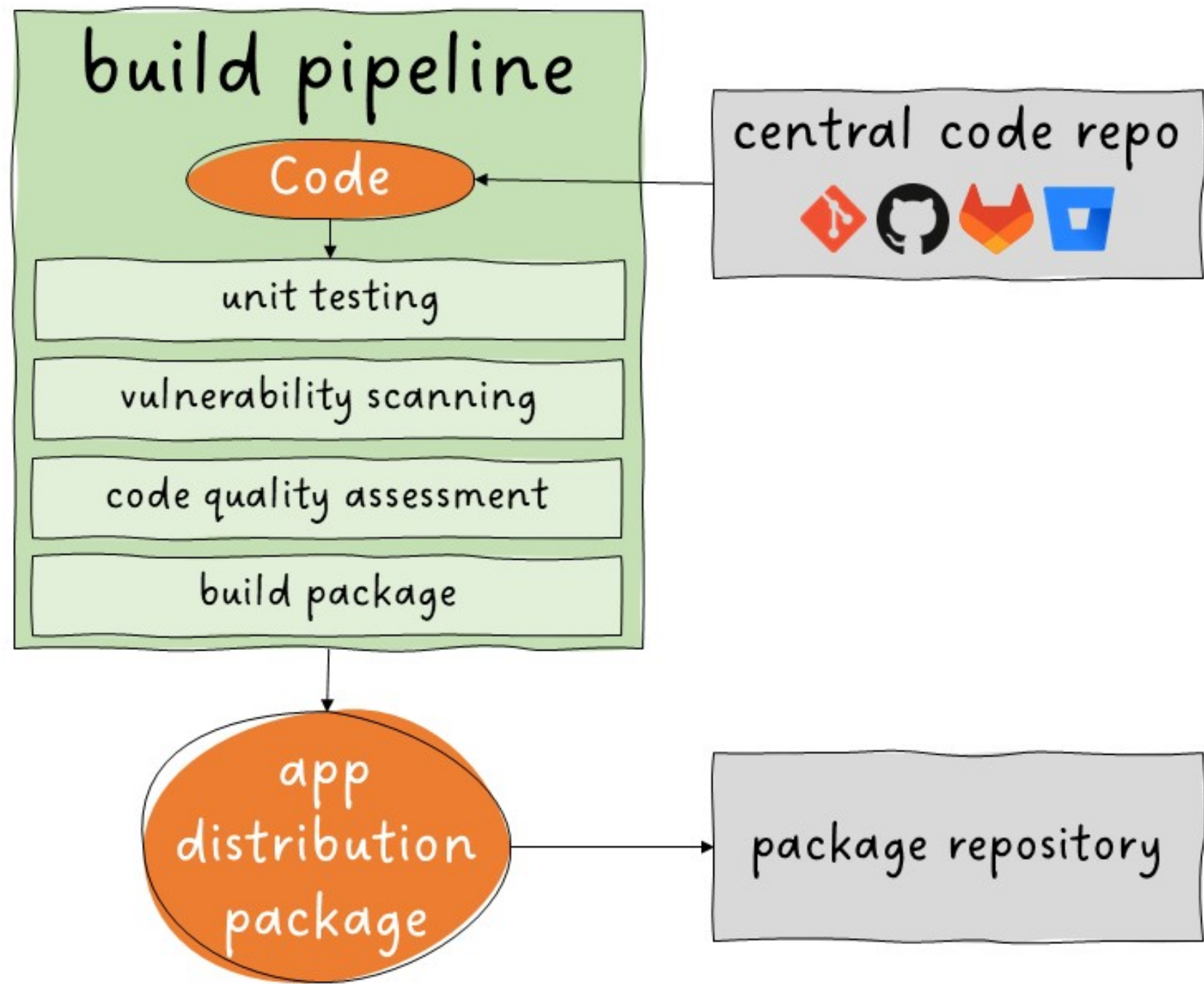




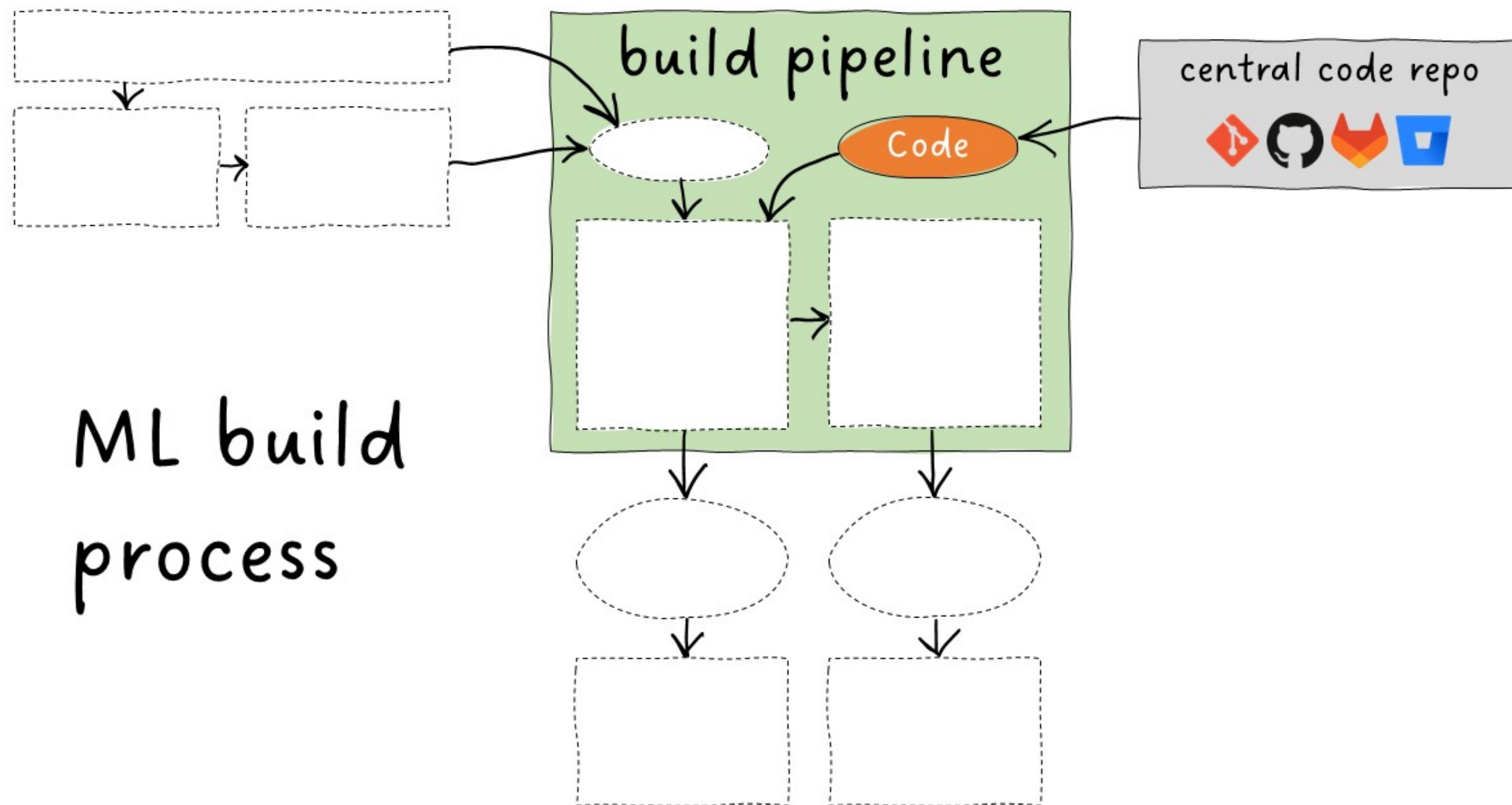




# App build process

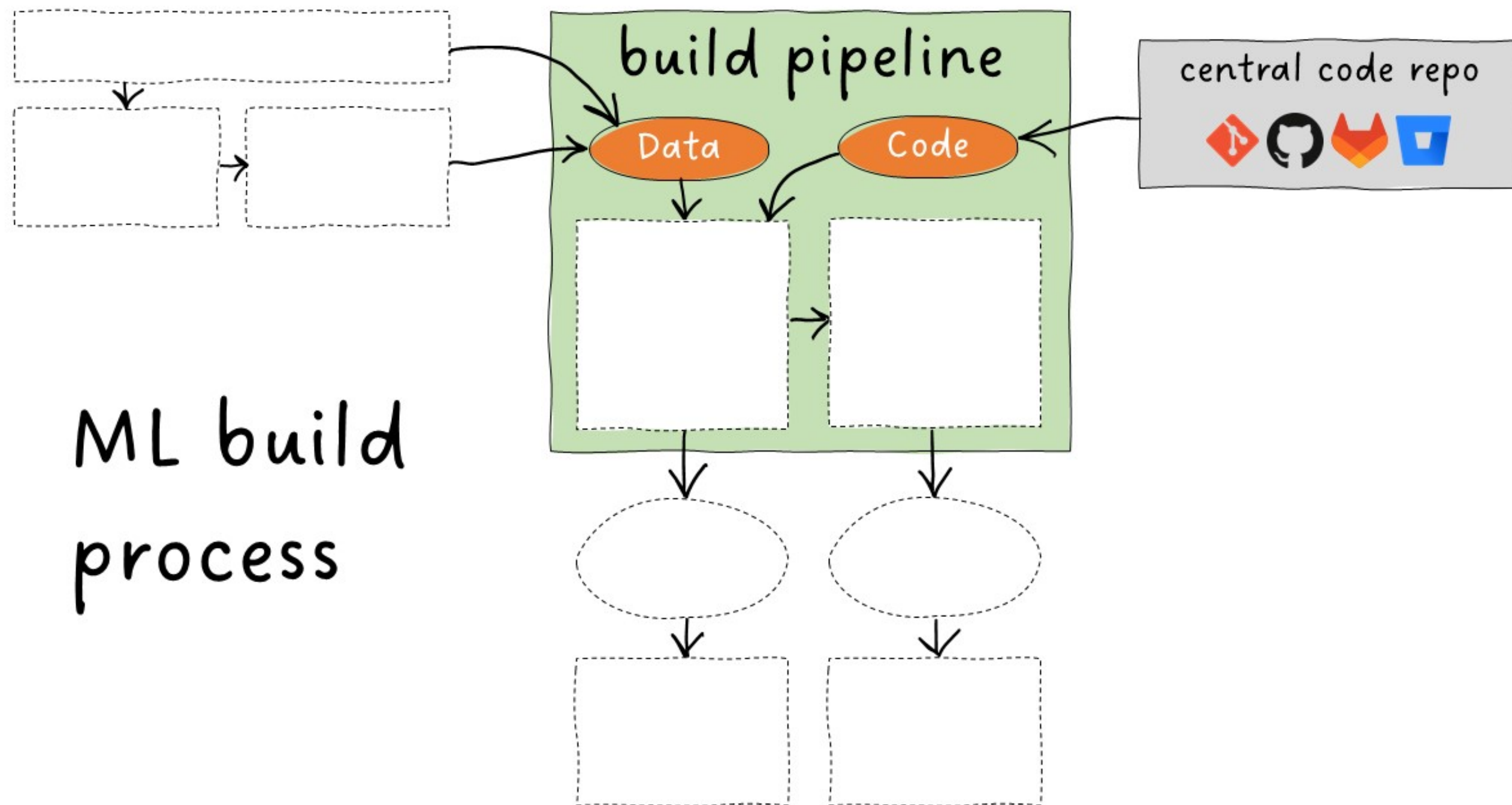




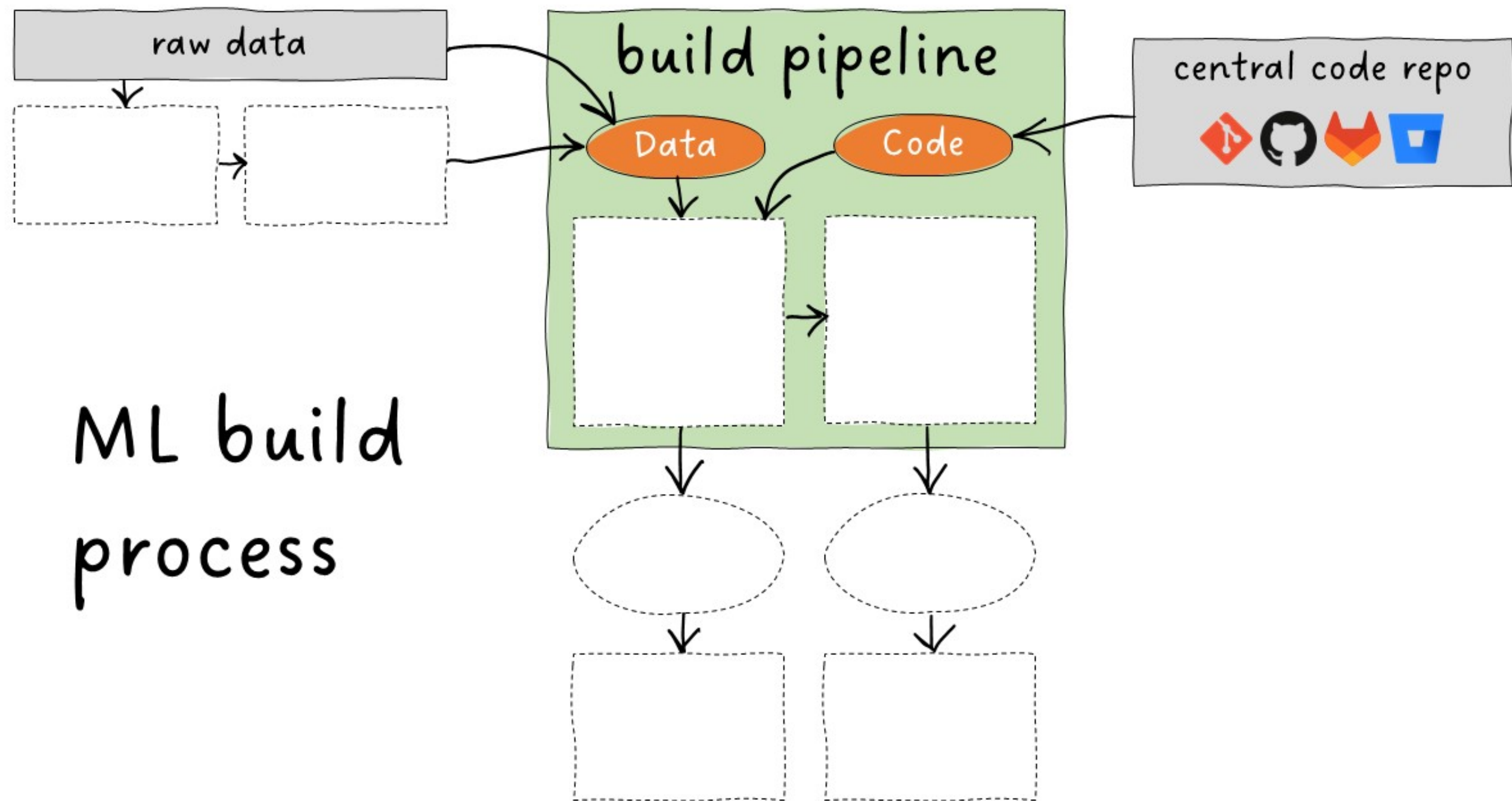


ML build  
process

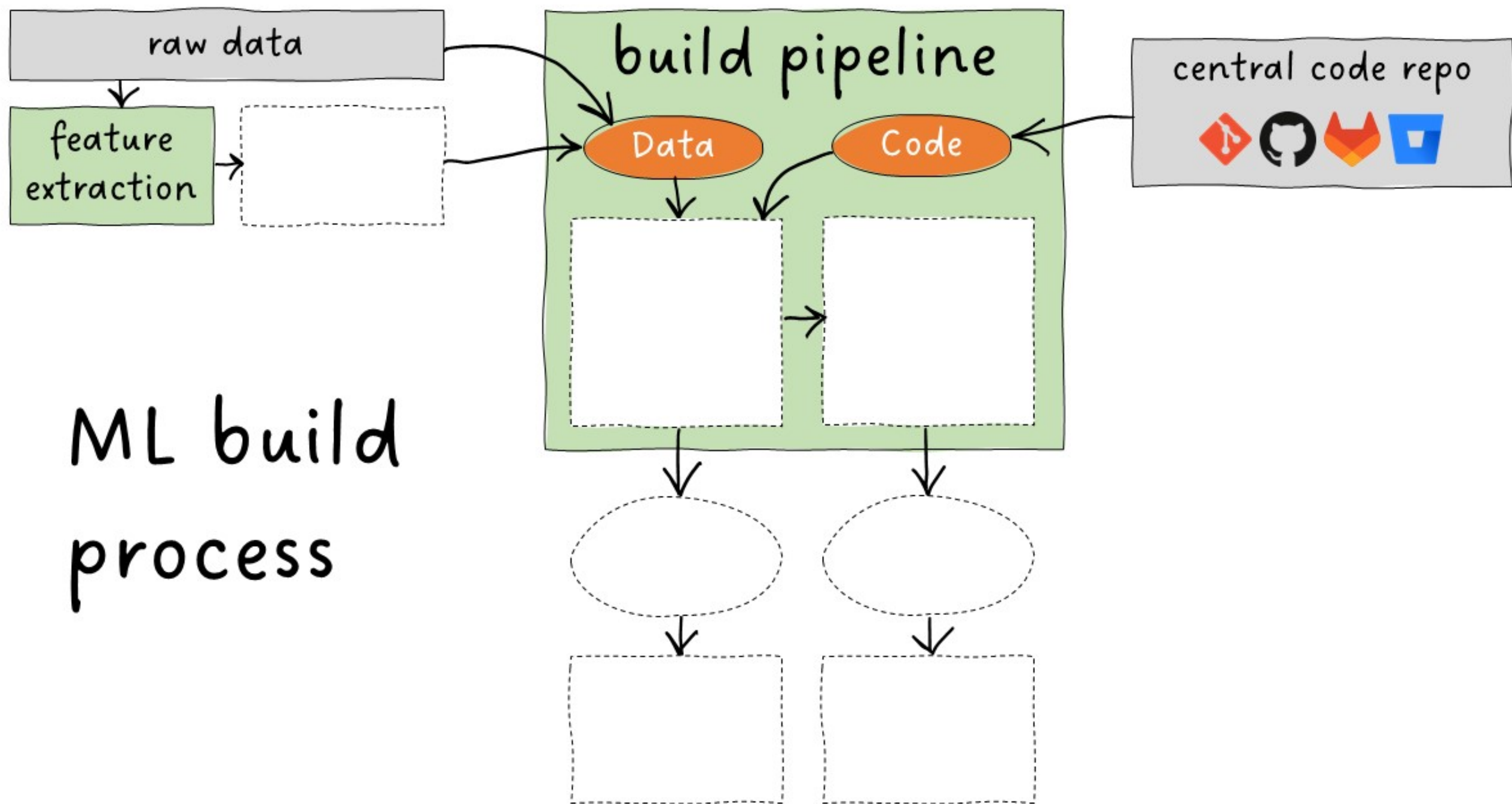


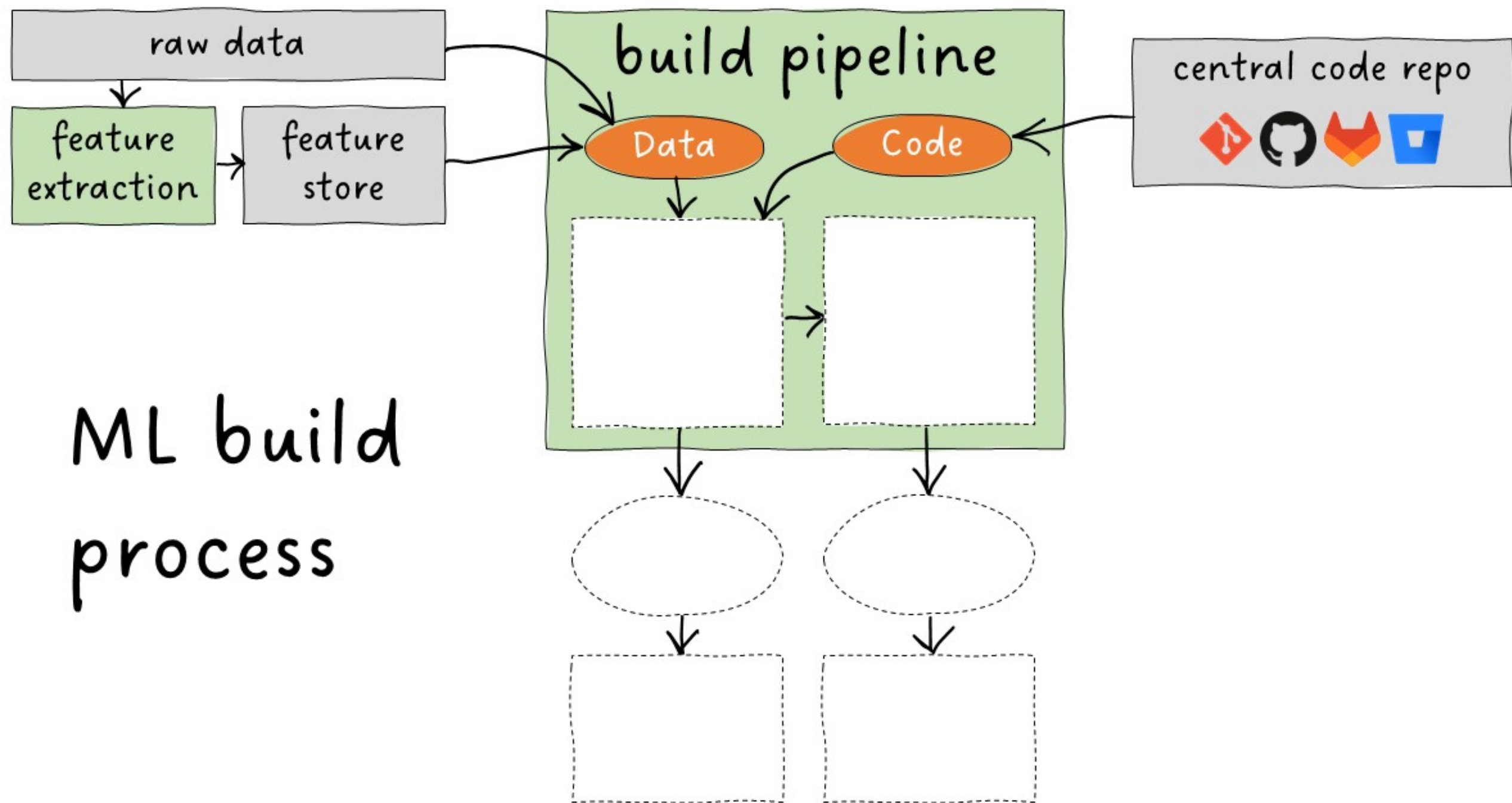


ML build  
process



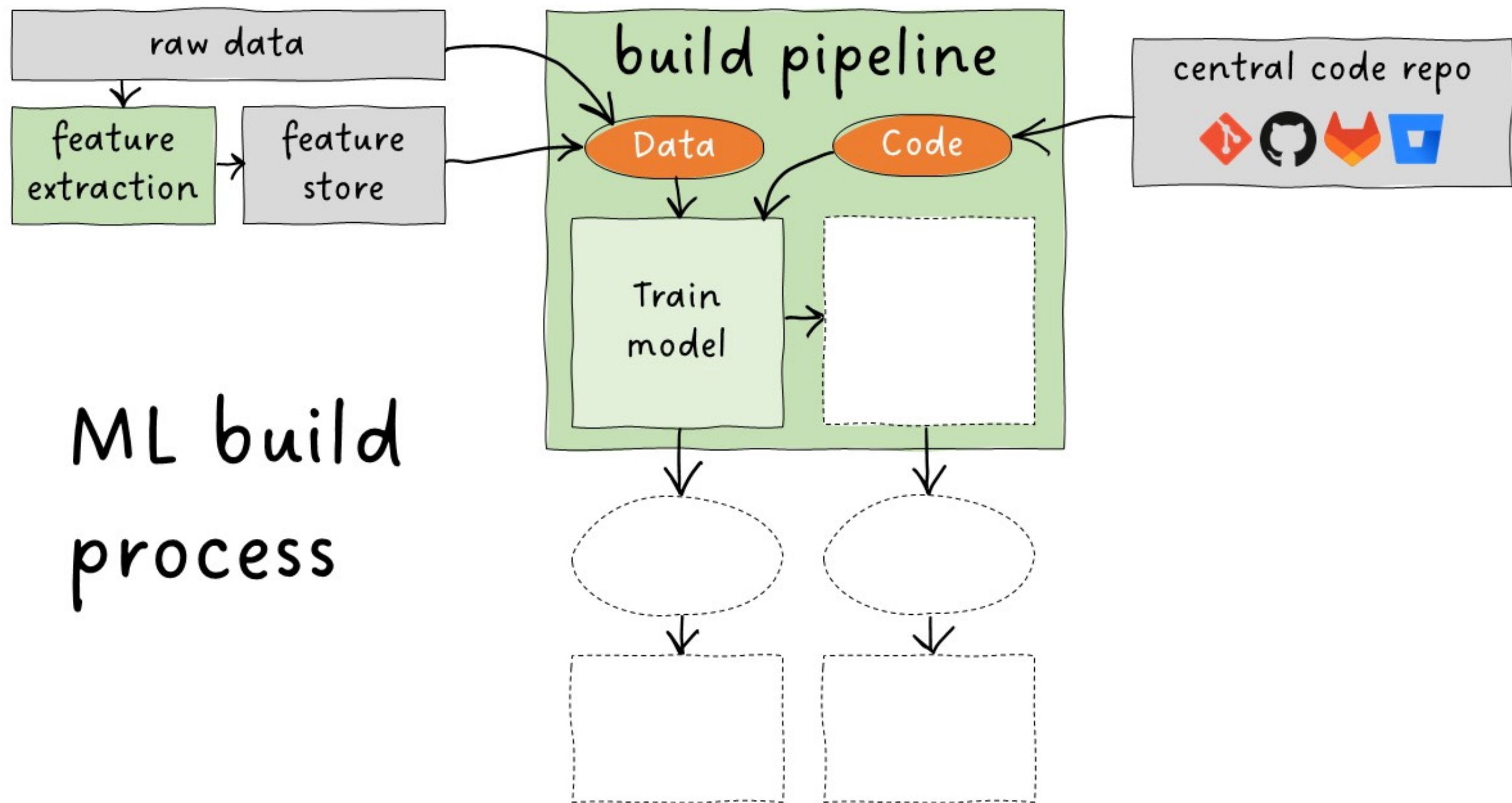
ML build  
process



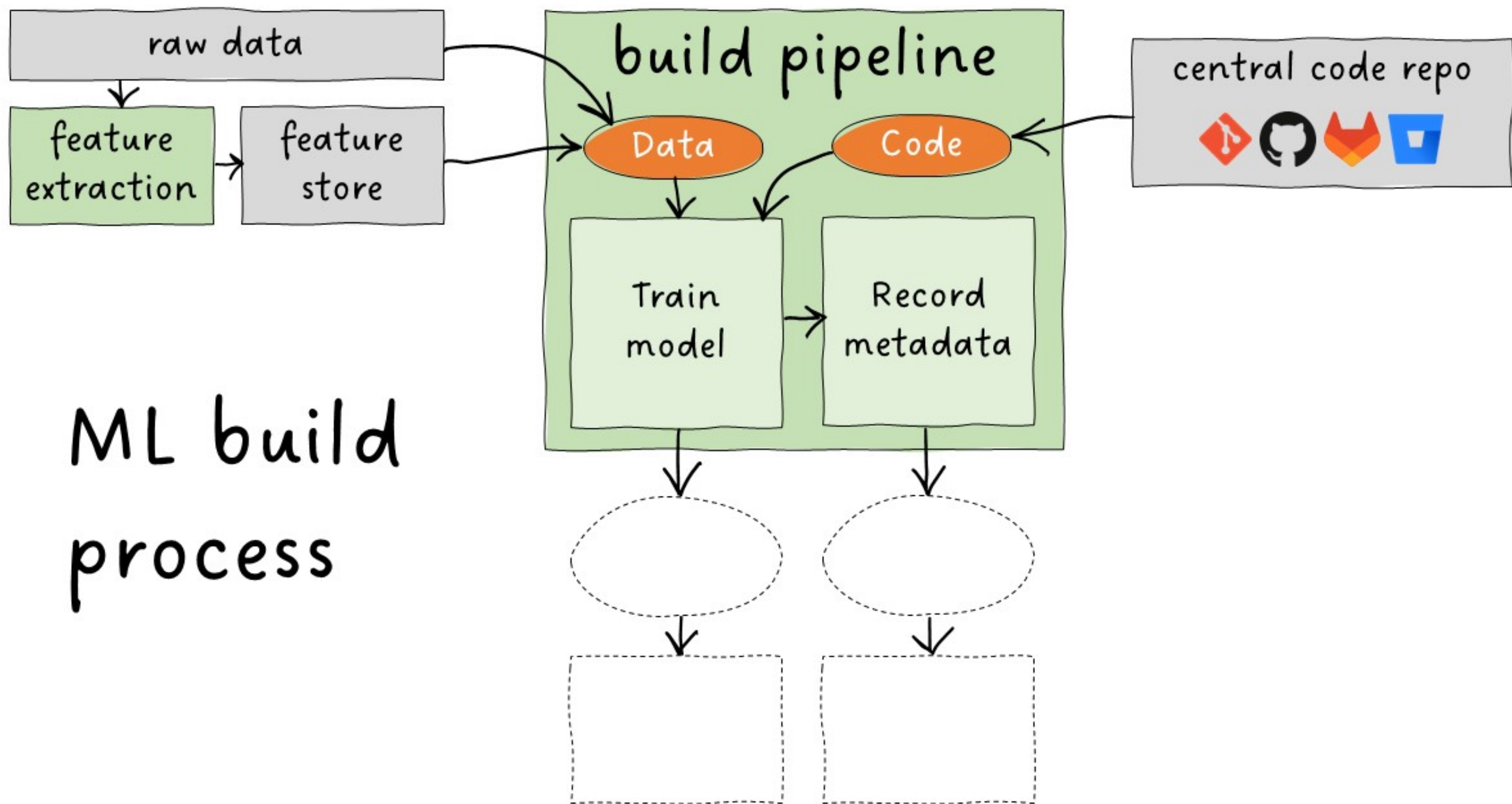


ML build  
process



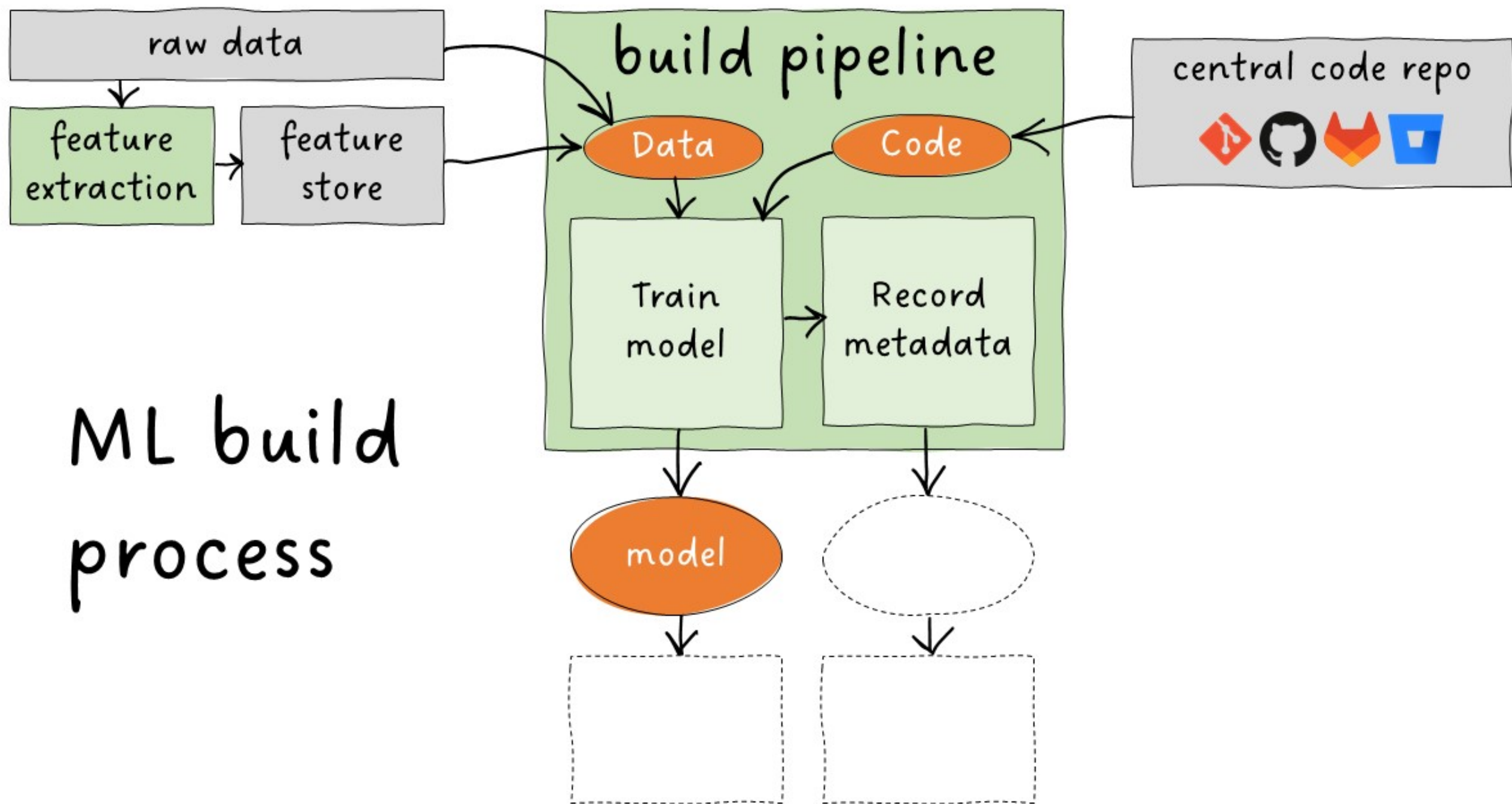


ML build  
process

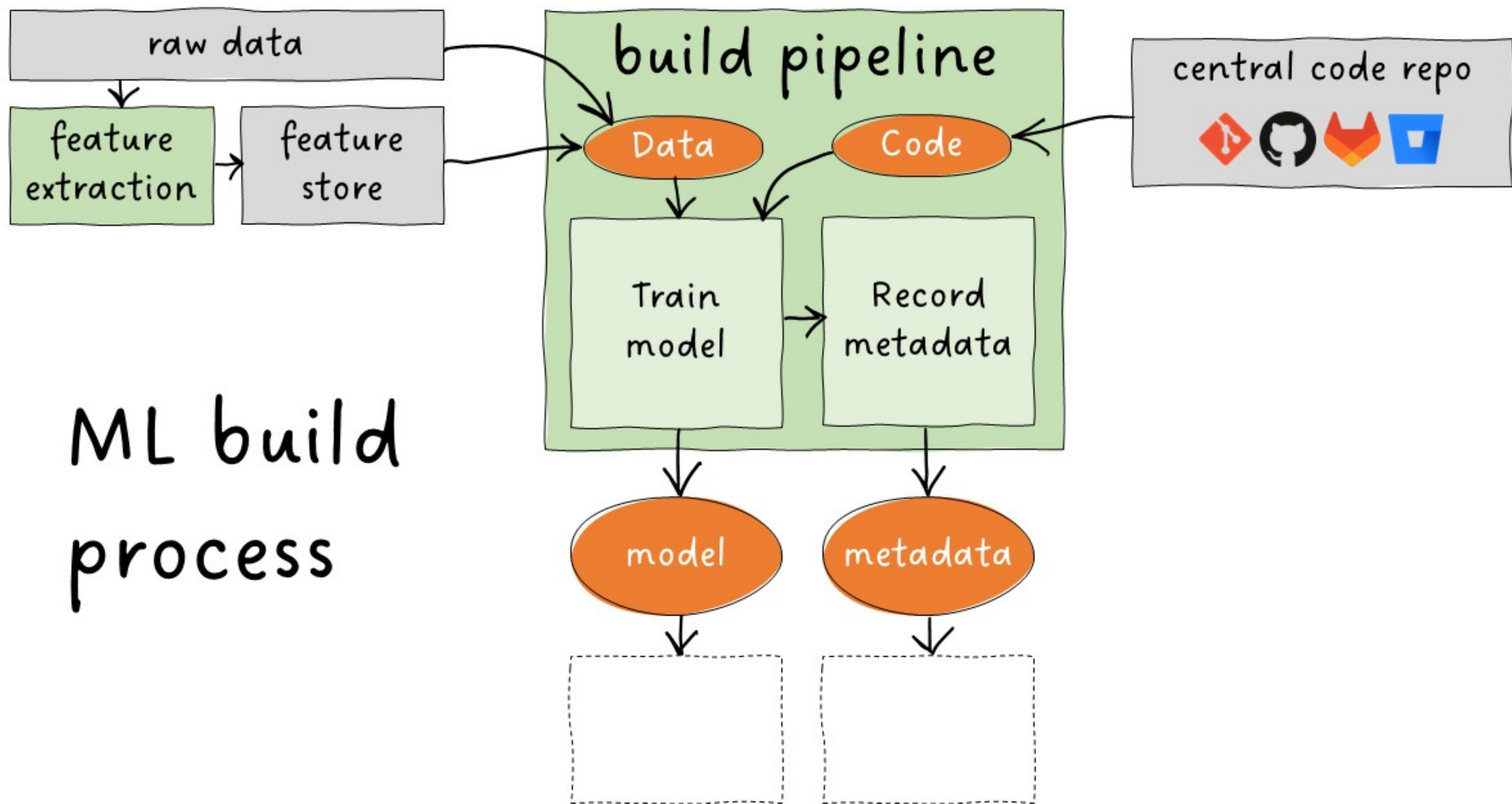


ML build  
process

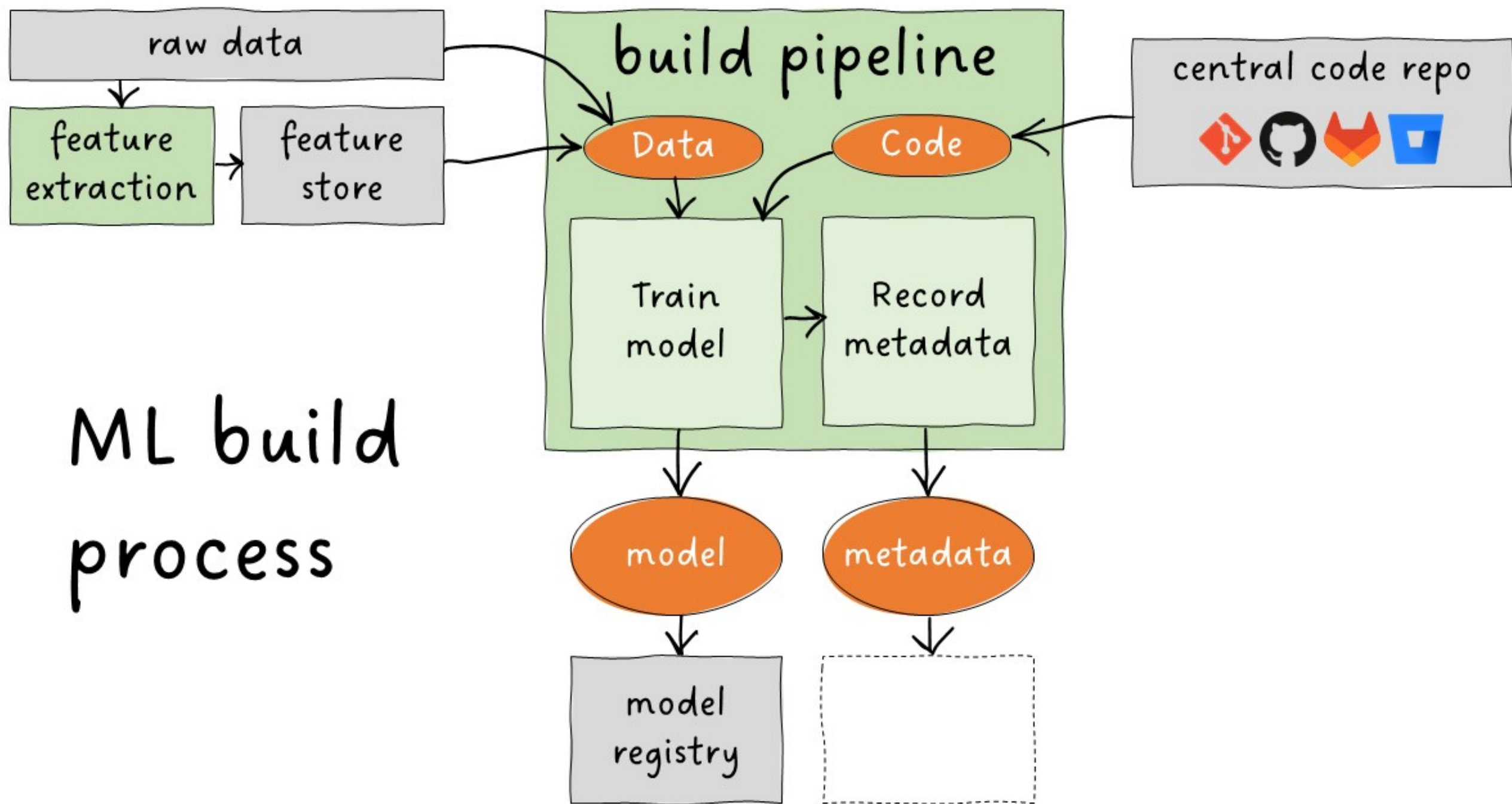


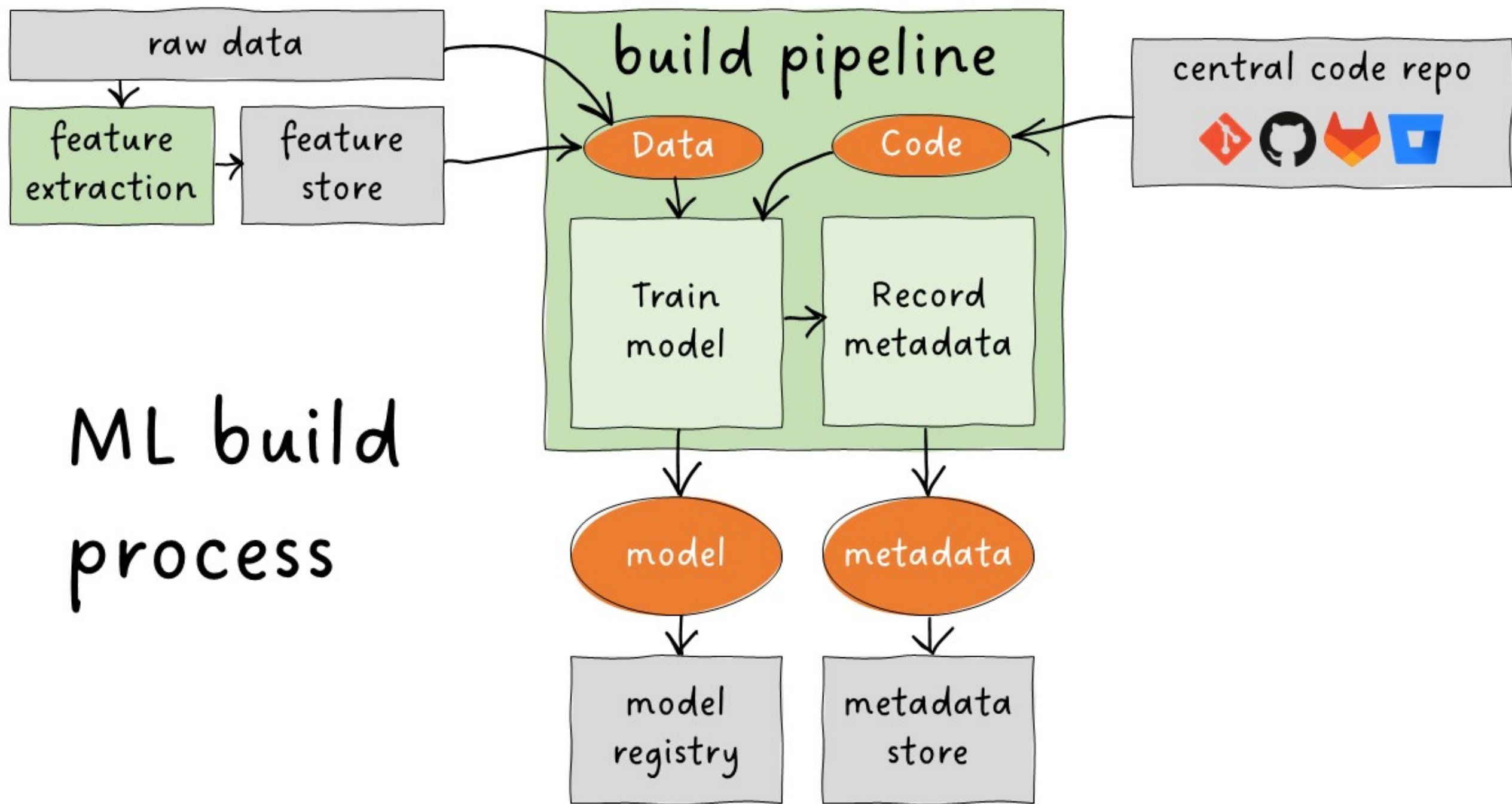


ML build  
process



ML build  
process



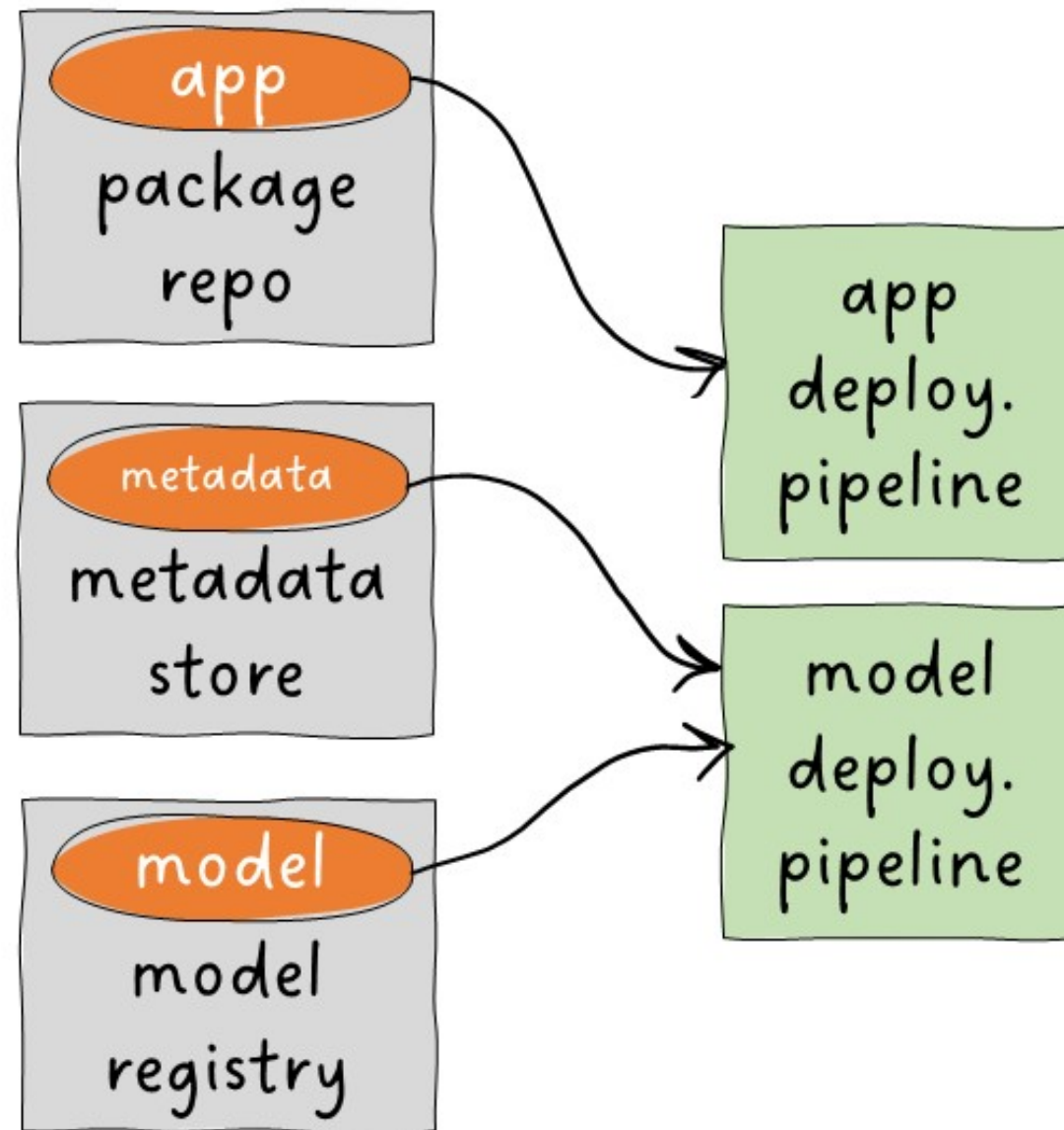


ML build  
process

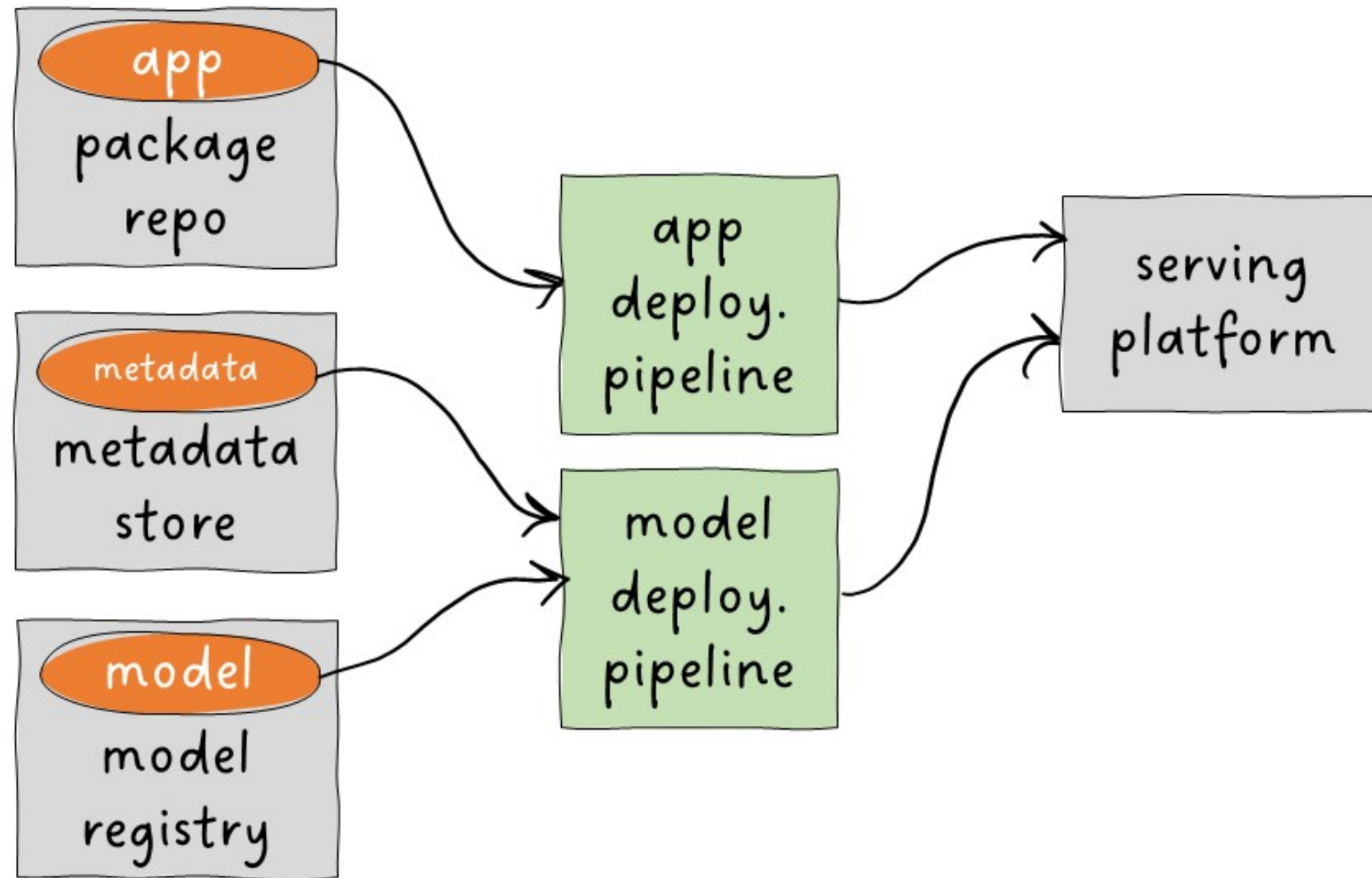


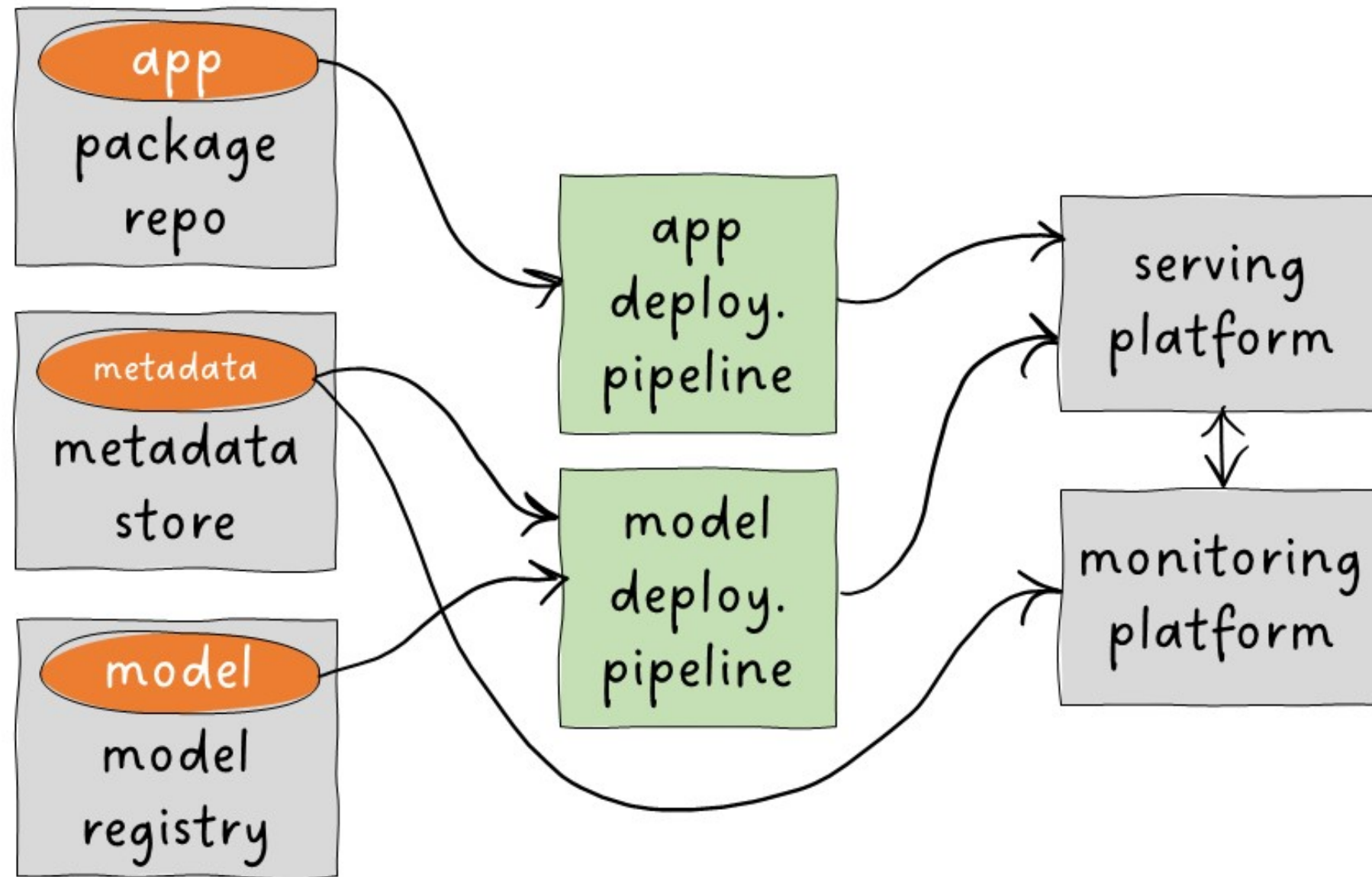
app  
deploy.  
pipeline

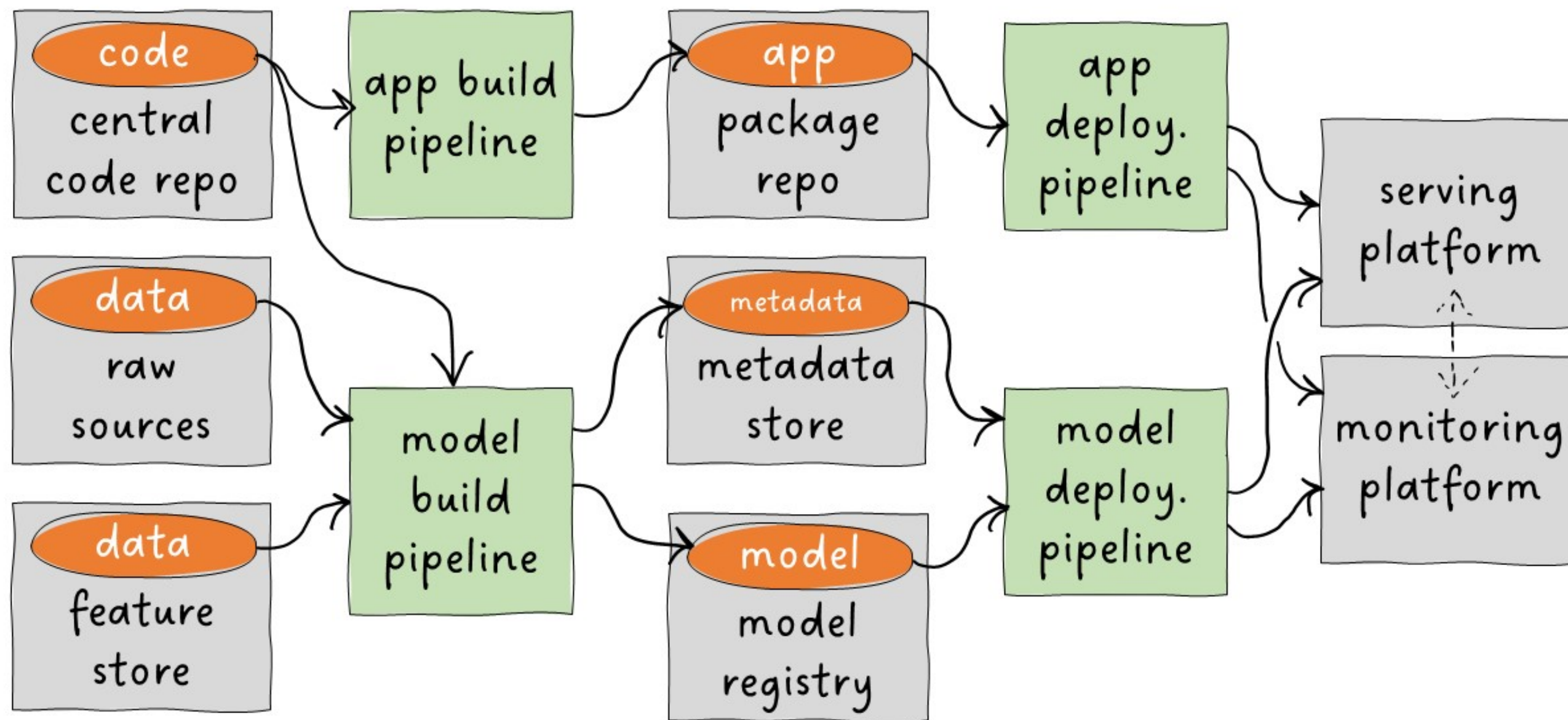
model  
deploy.  
pipeline











# Let's practice!

MLOPS DEPLOYMENT AND LIFE CYCLING